

Chaos in Large Language Models

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Introduction and Motivation

LLMs are highly non-linear systems. They are Deep Neural Networks trained on large amounts of data.

There is growing interest in interpretability and understanding the model's dynamics.

Reproducibility and determinism is important in order to do Reinforcement Learning on-policy. Quantifying sensitivity to perturbations and initial conditions is necessary.

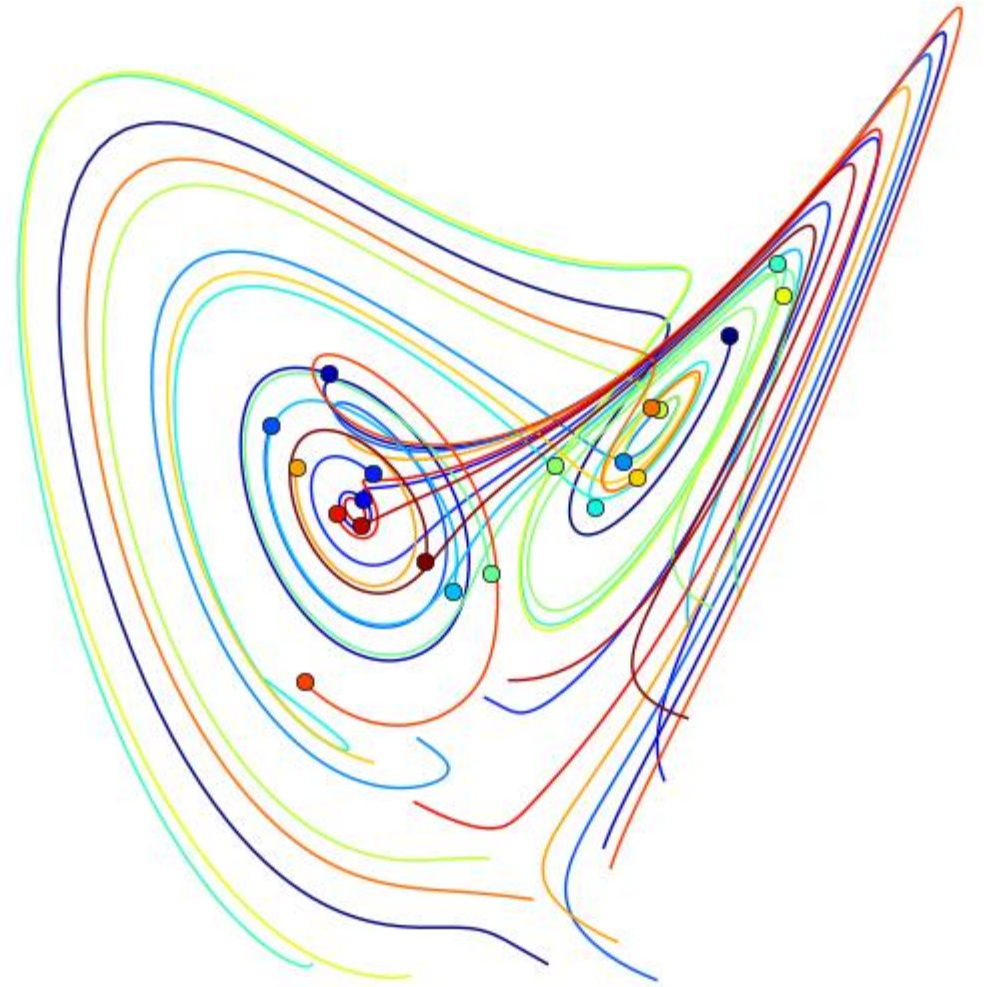
They are an interesting case study for chaos theory.

Typical chaotic system: Lorenz

$$\frac{dx}{dt} = \sigma(y - x),$$

$$\frac{dy}{dt} = x(\rho - z) - y,$$

$$\frac{dz}{dt} = xy - \beta z.$$



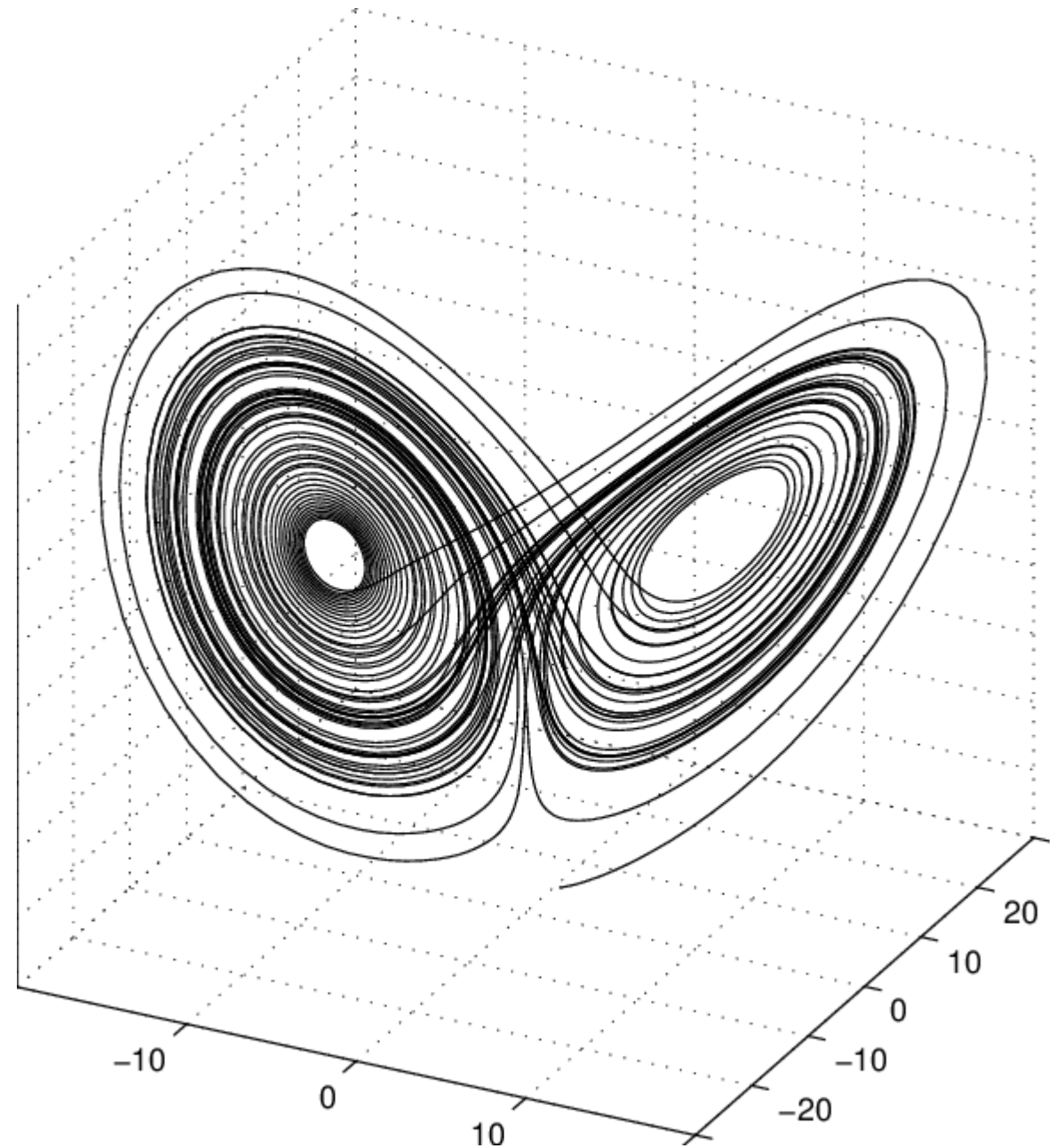
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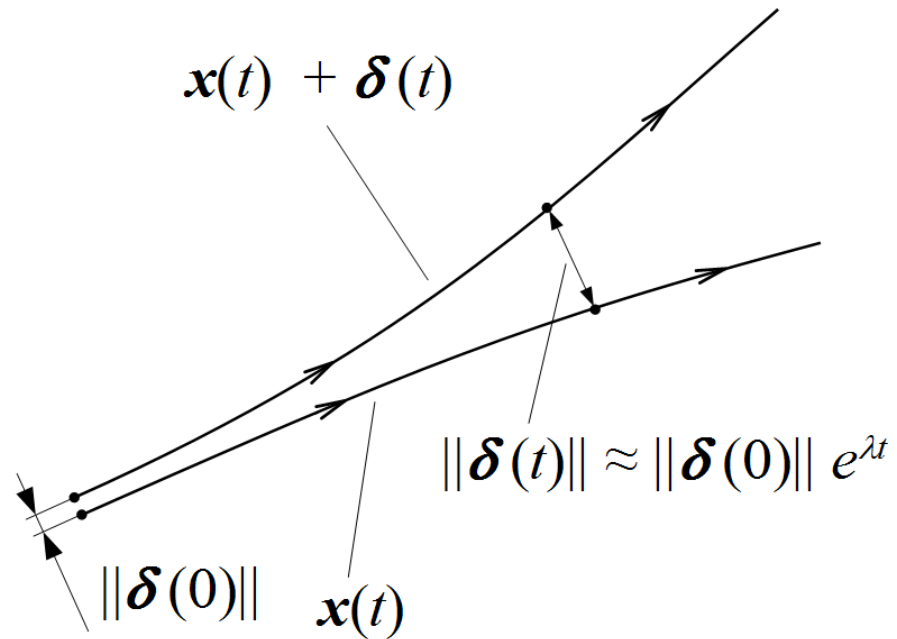
$$\frac{dy}{dt} = x(\rho - z) - y,$$

$$\frac{dz}{dt} = \underline{xy} - \beta z.$$

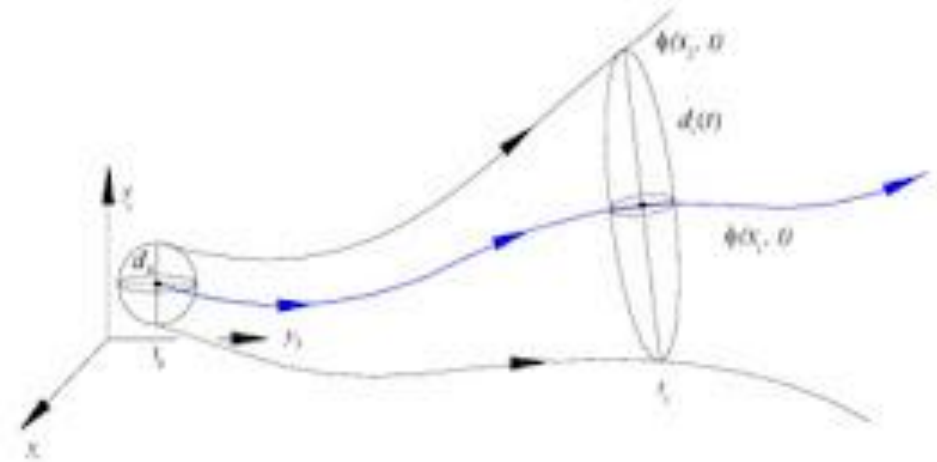
Nonlinear coupling of
variables



Divergence of trajectories. Predictability?



Lyapunov Exponent

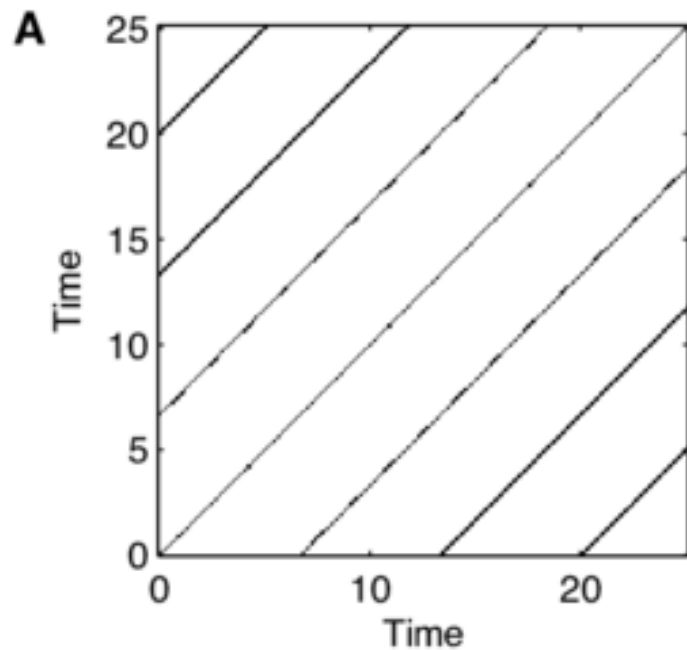


- $|\delta Z(t)| = e^{\lambda t} |\delta Z(0)|$

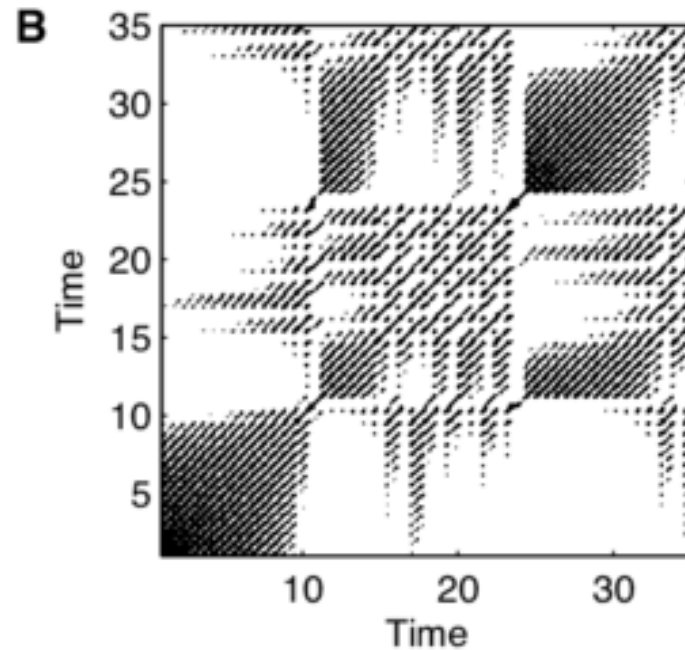
RECURRENCE PLOTS

$$R_{i,j} = \Theta(\varepsilon_i - \|\vec{x}_i - \vec{x}_j\|),$$

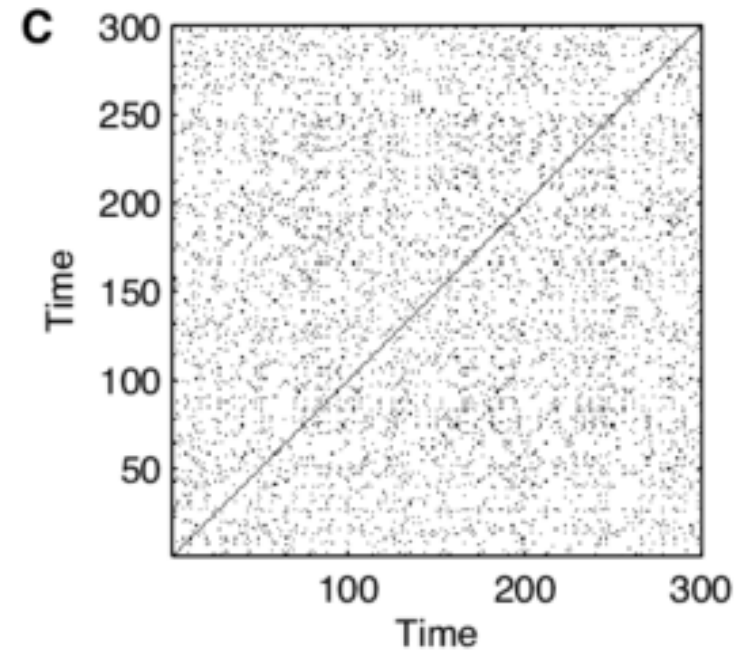
Periodic sine



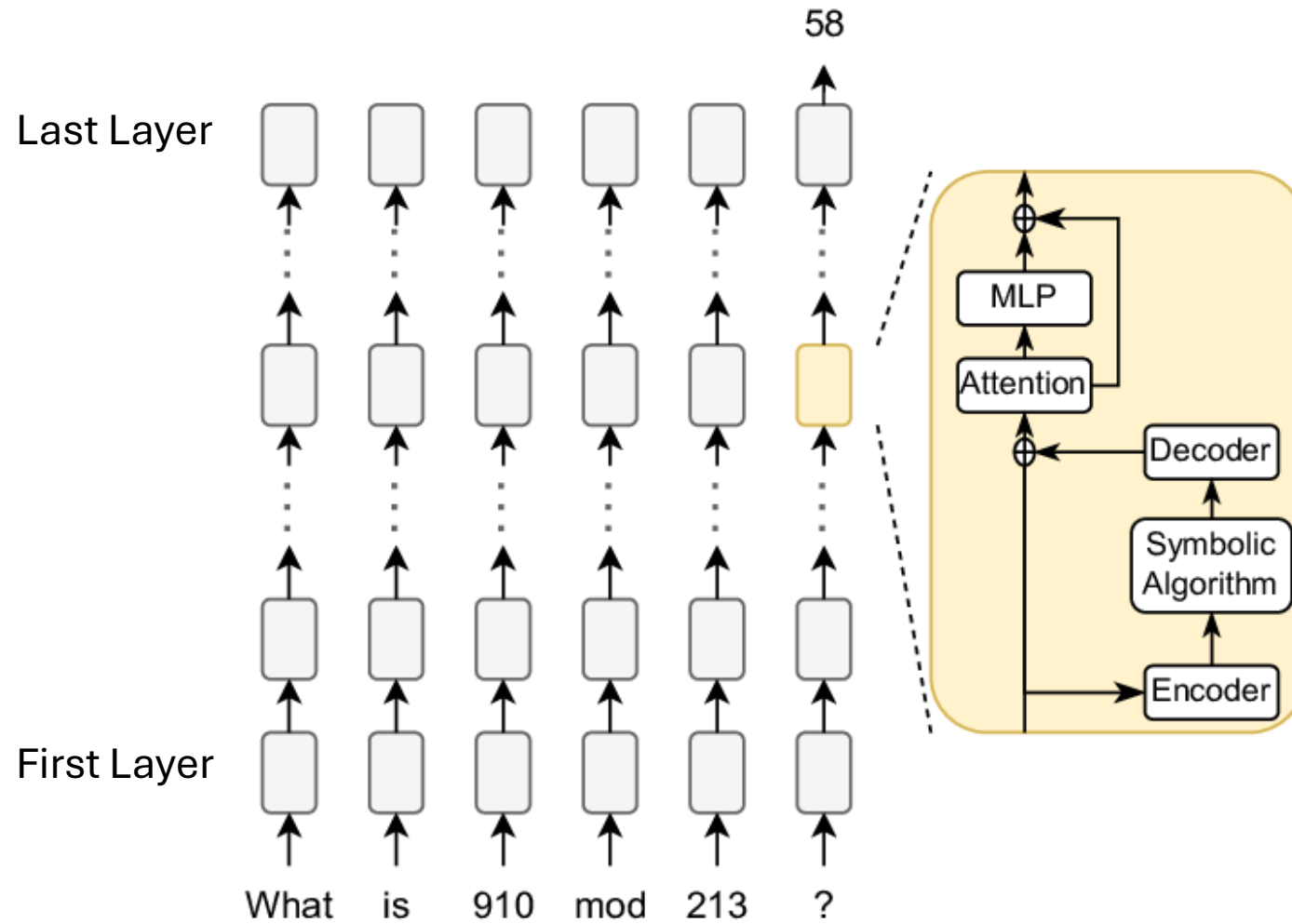
chaotic



random



LLMs



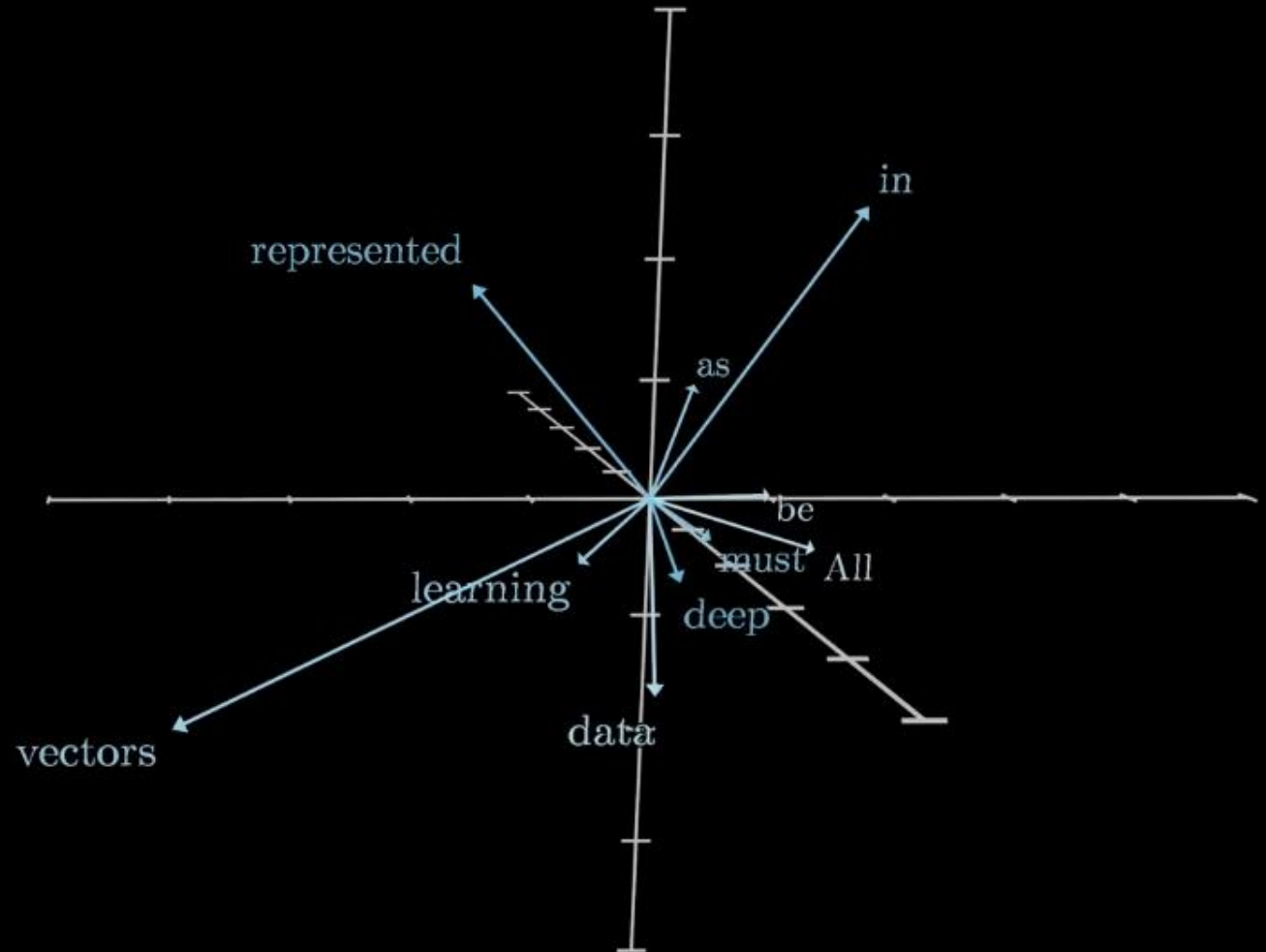
Words

“Embedding”

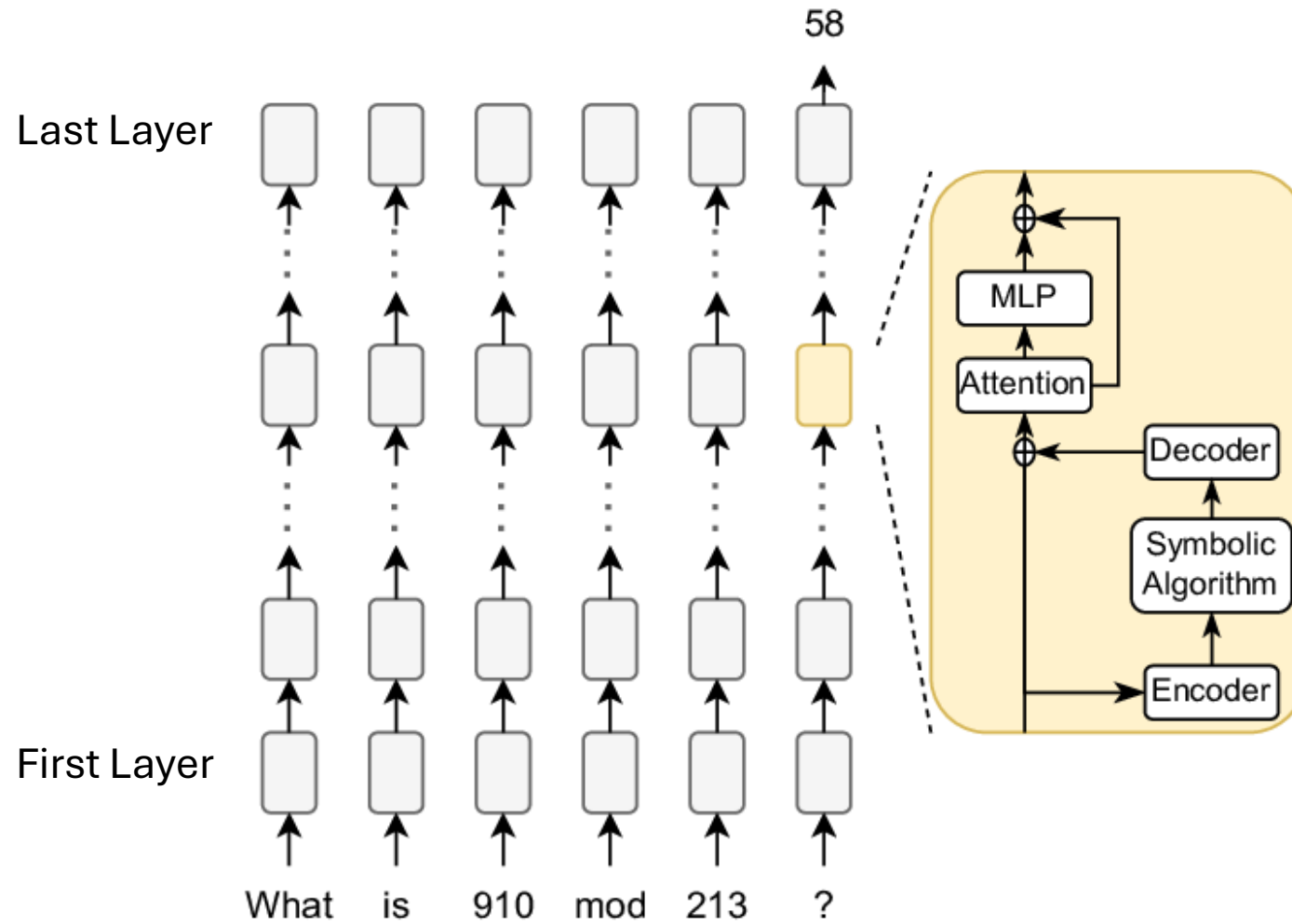


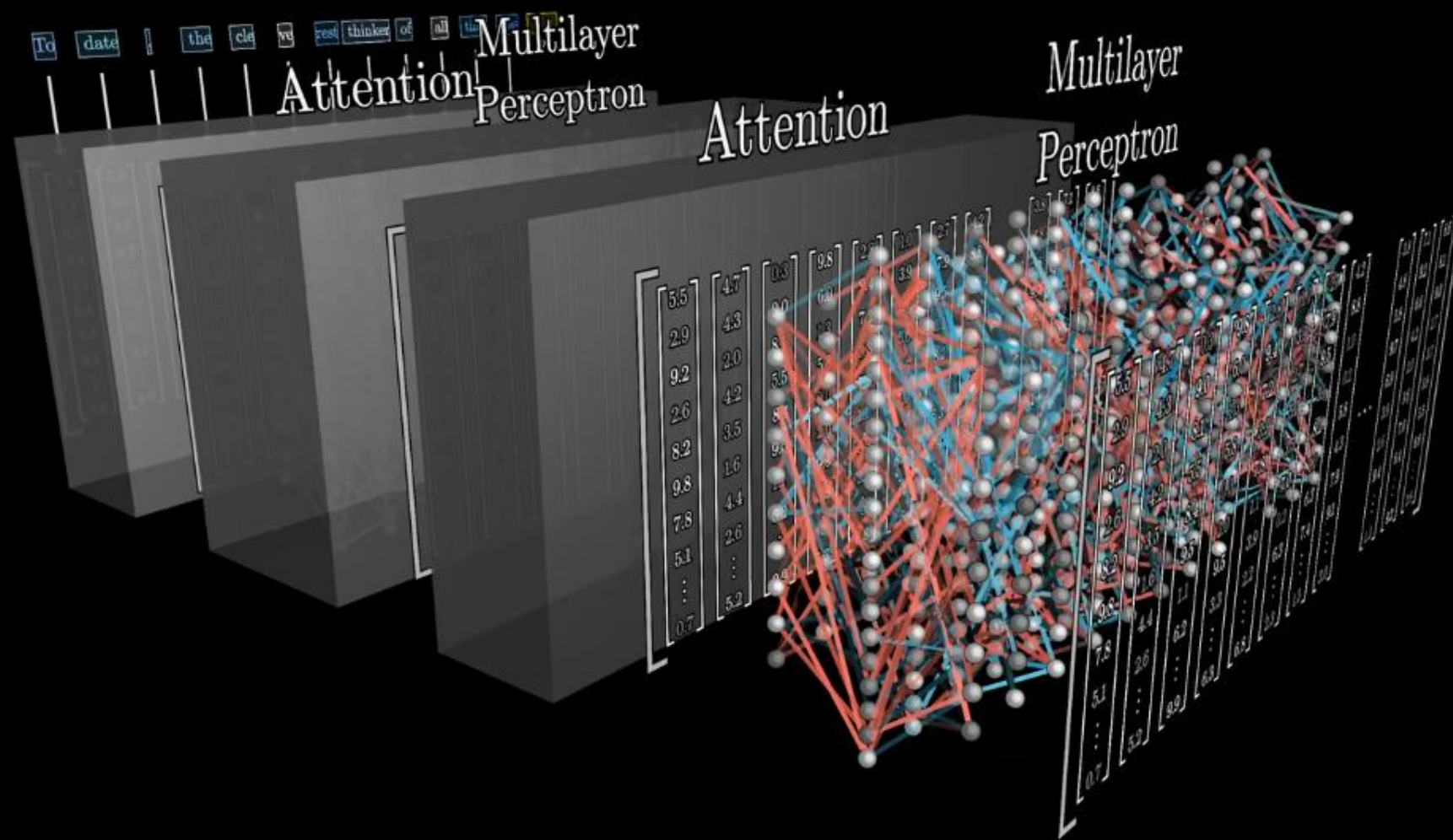
Vectors

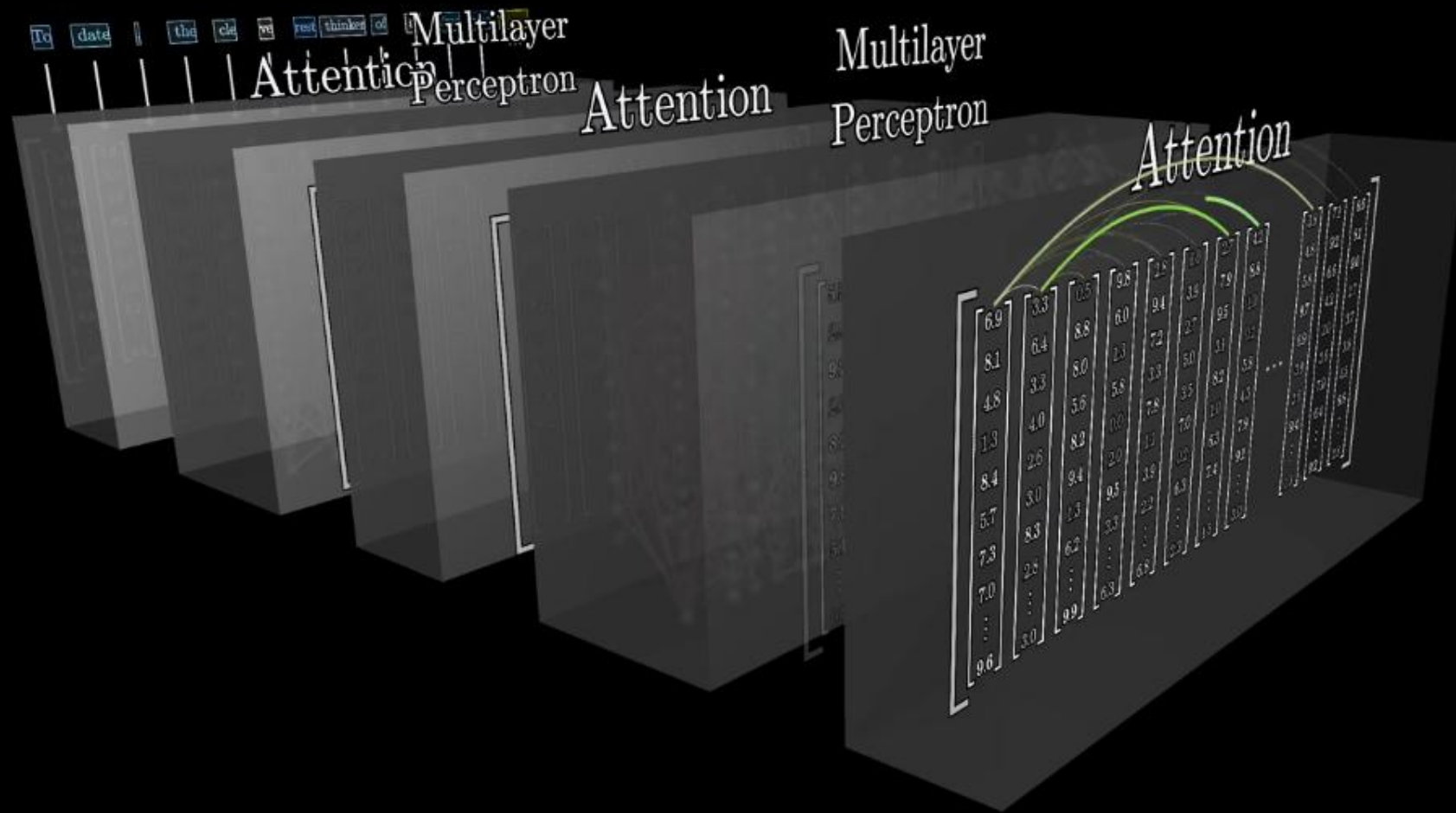
All
data
in
deep
learning
must
be
represented
as
vectors



LLMs

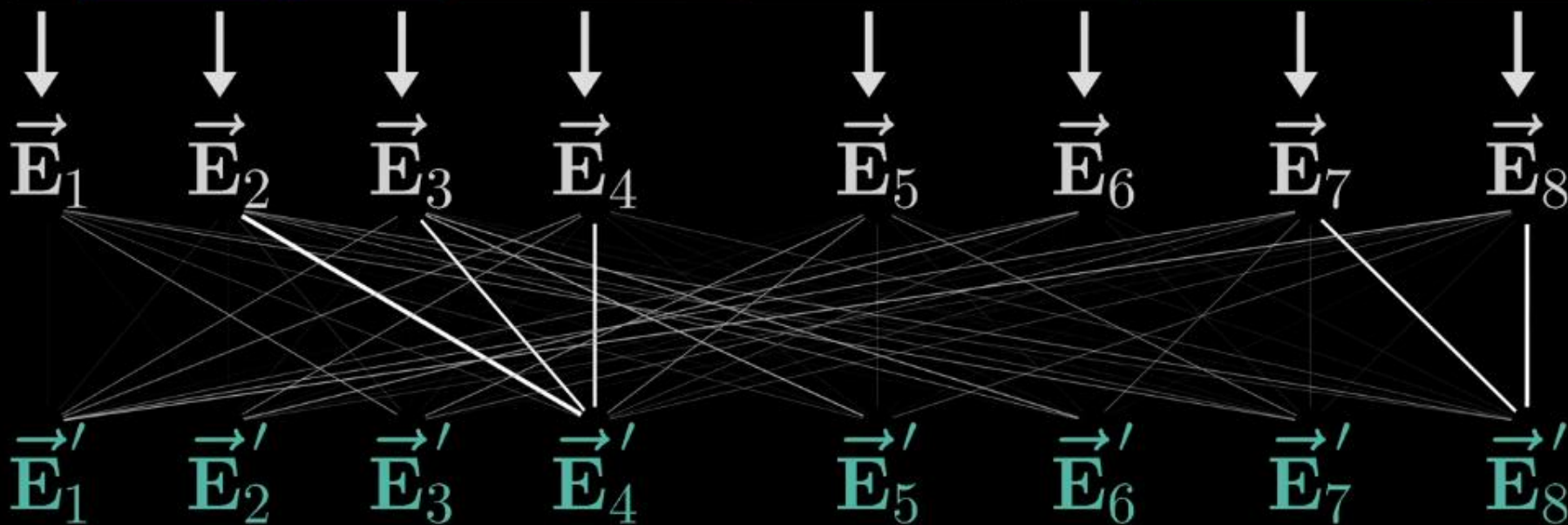




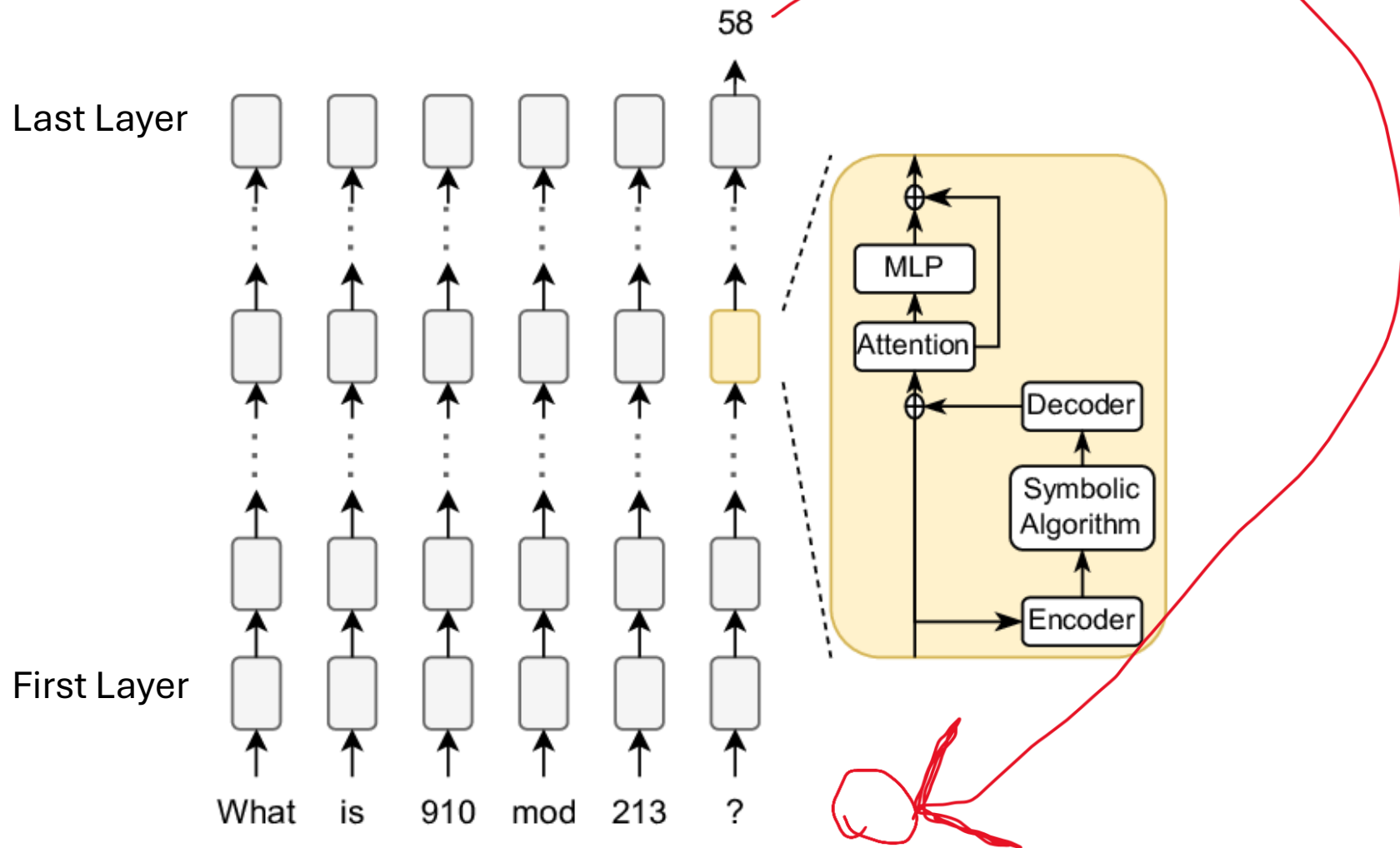




a fluffy blue creature roamed the verdant forest

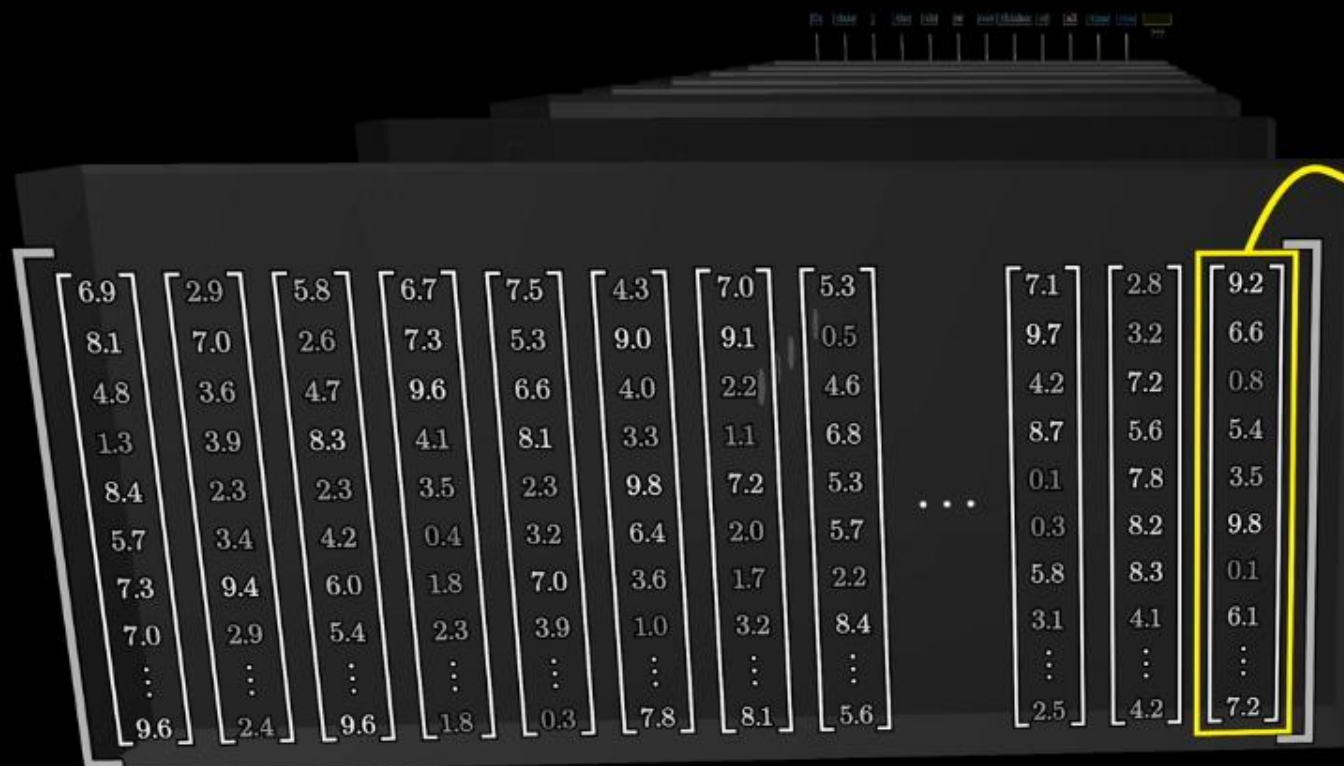


LLMs



To date, the cleverest thinker of all time was

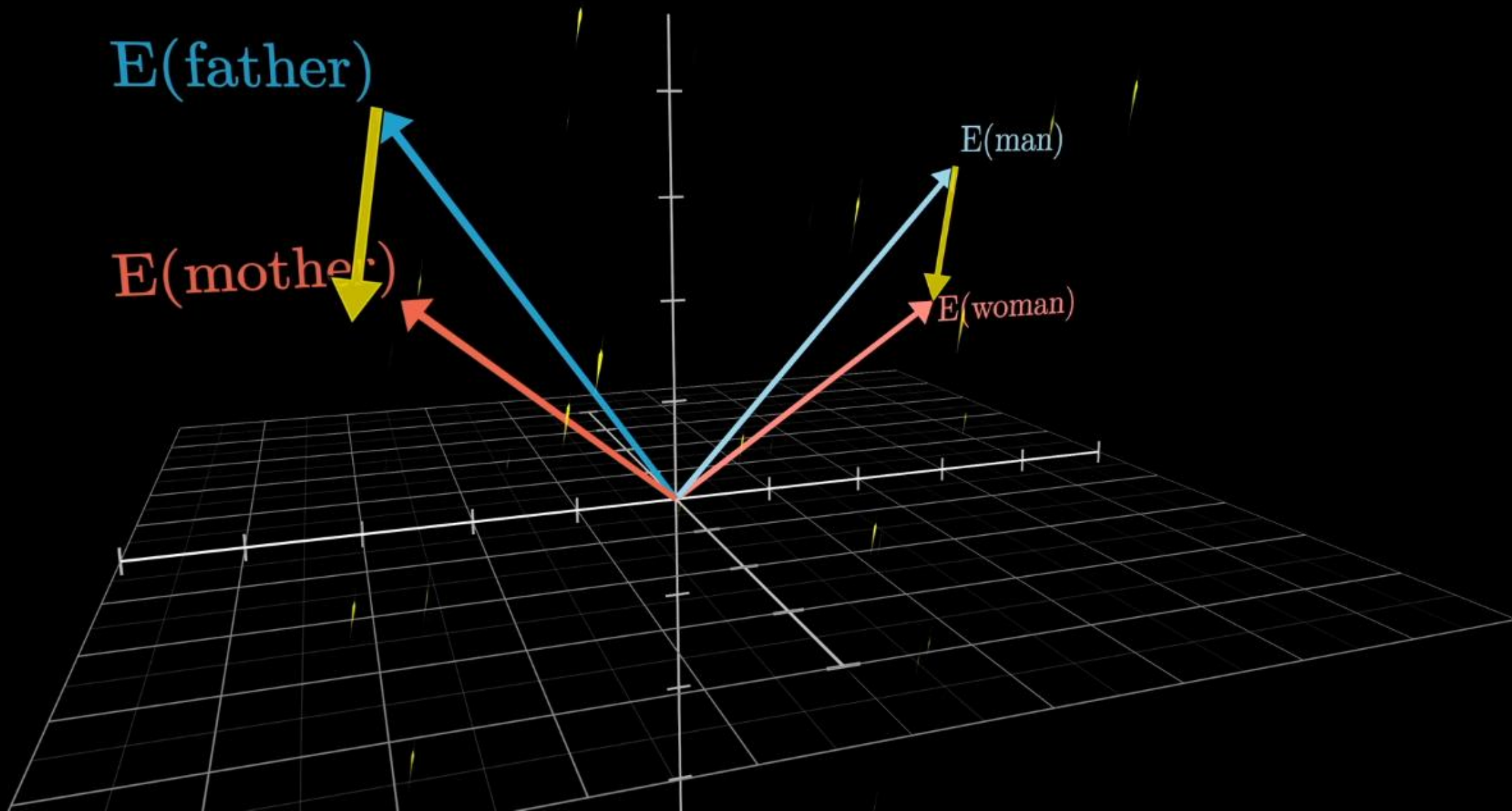
???



the 8.82%
probably 4.37%
John 4.04%
Sir 3.66%
Albert 3.63%
Ber 3.31%
a 2.90%
Isaac 2.01%
undoubtedly 1.58%
arguably 1.33%
Im 1.16%
Einstein 1.13%
Ludwig 1.04%
⋮



$$E(\text{mother}) - E(\text{father}) \approx E(\text{woman}) - E(\text{man})$$



Chaos and LLMs

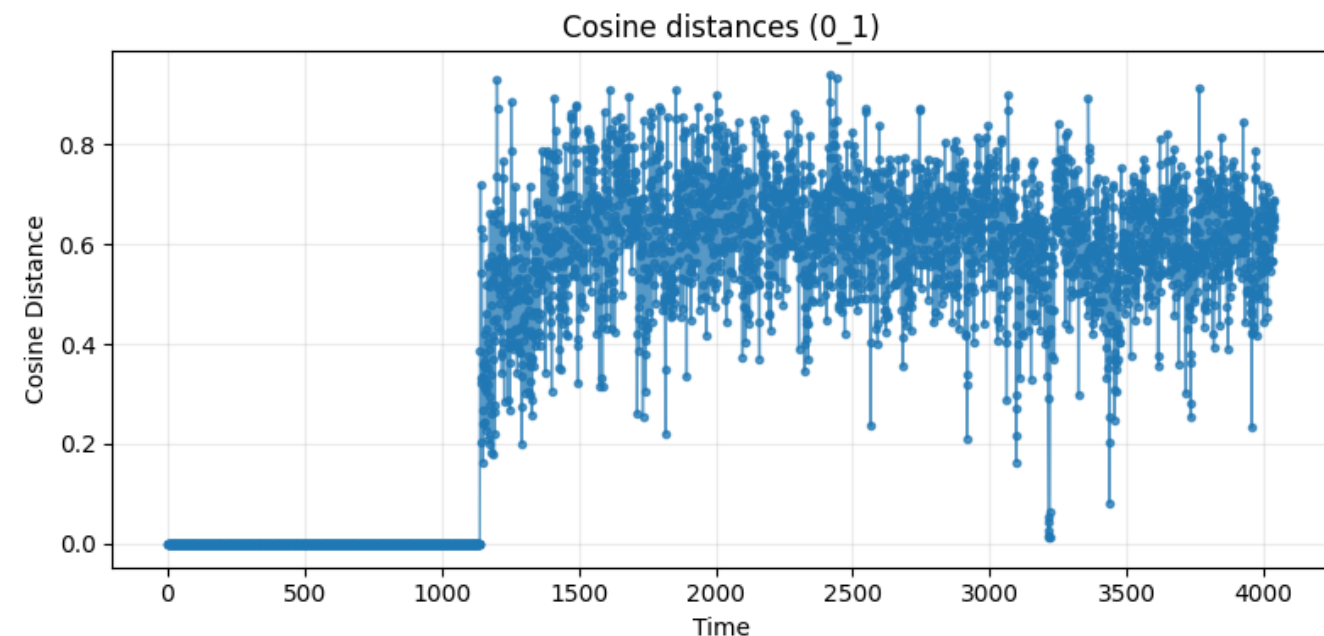
Adapt concepts from chaos theory to LLMs

Let's look at their similarities, generate data with LLM and analyze with chaos tools

Chaotic systems	LLM
Phase Space	Hidden State Space
Phase Space Trajectory	Sequence of Token Text or Hidden States
Differential Equation	Model
Numerical Solution	Forward Pass Inference
“blob” of initial conditions	Tokenize and embed initial prompt. Then add random tensor (small magnitude)
Determinism	Greedy sampling

Challenges:

- Chaos metrics didn't work out of the box
- There is no canonical distance metric
- Hidden states are very high dimensional

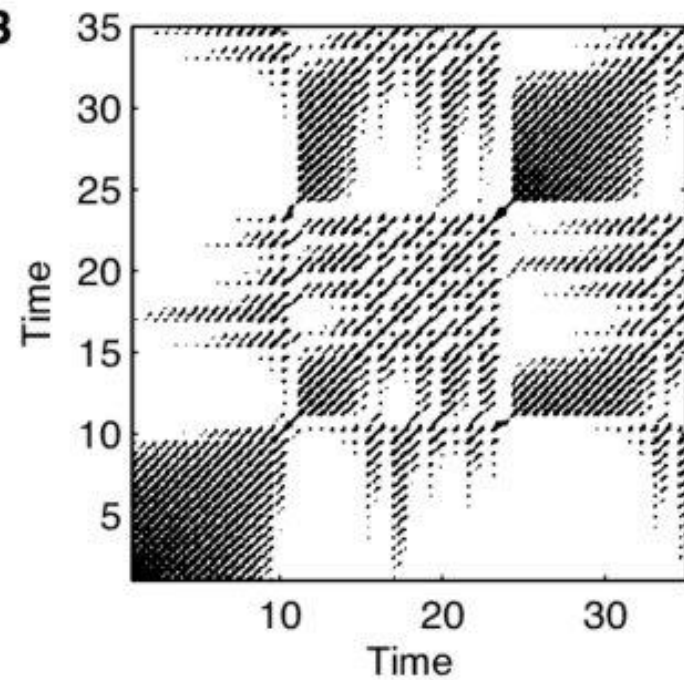


Divergence between two trajectories

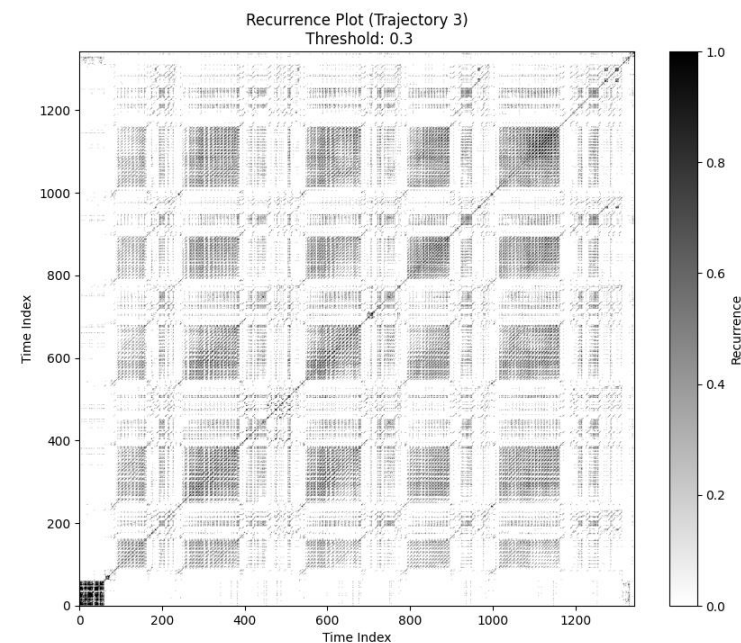
First results:

- Noisy
- Distance metric did not correspond to semantic distance

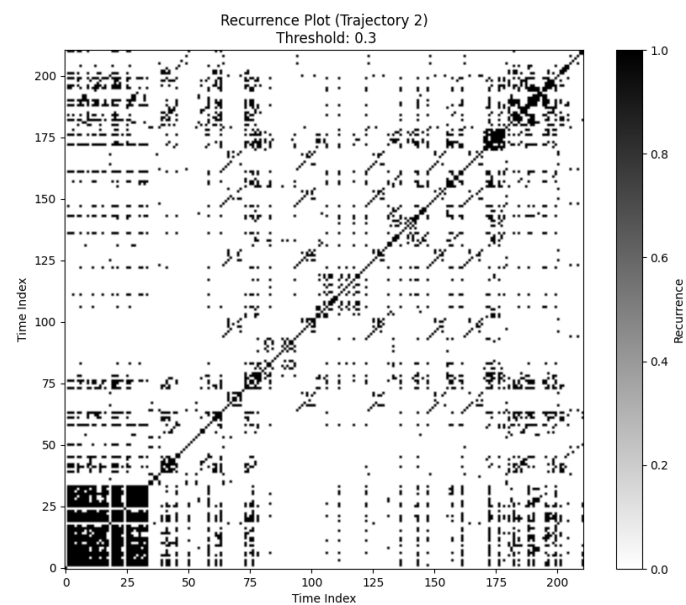
Lorenz



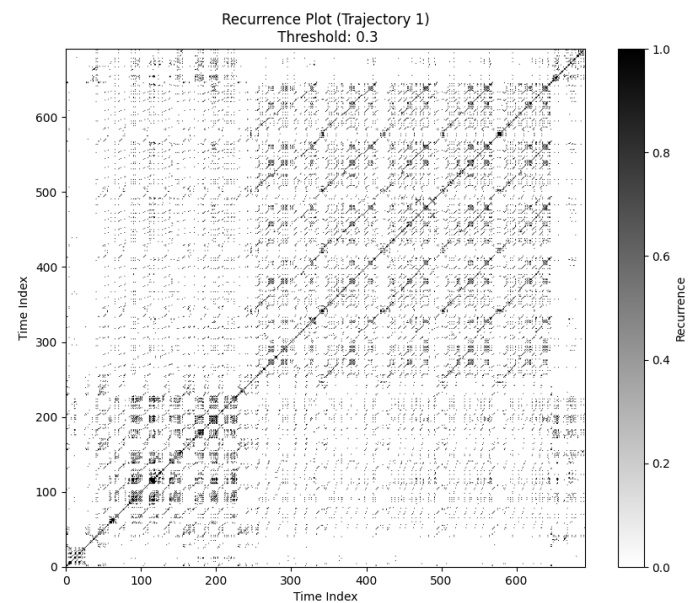
LLM
(Last Layer)



LLM
(Last Layer)

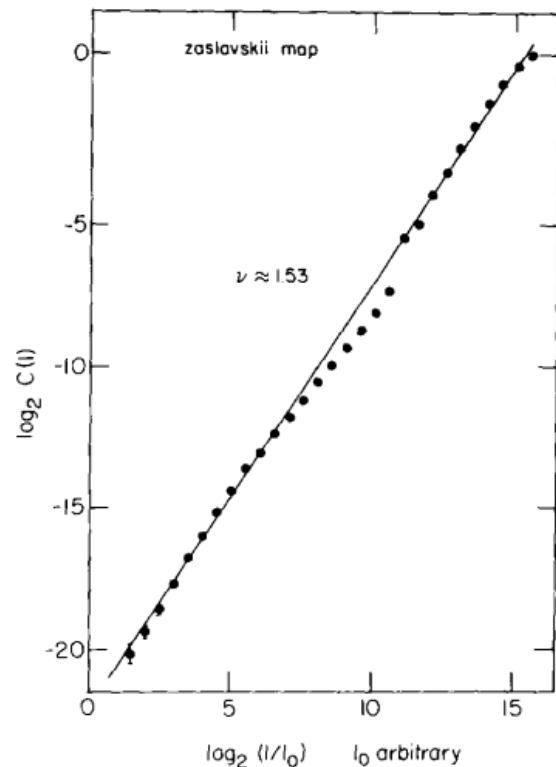


LLM
(Last Layer)

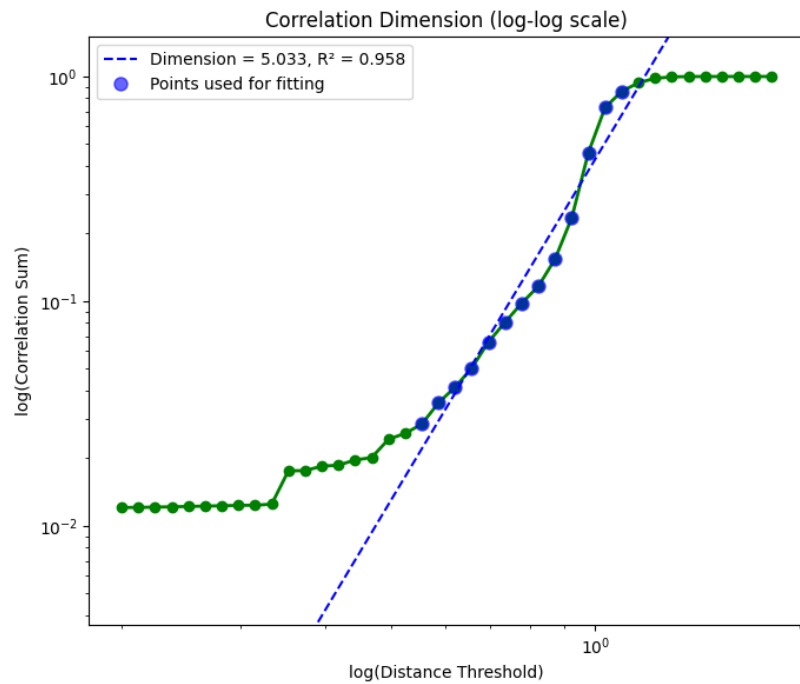


Correlation Dimension

$$C(\varepsilon) = \lim_{N \rightarrow \infty} \frac{1}{N(N-1)} \sum_{i \neq j} \Theta(\varepsilon - \|\mathbf{x}_i - \mathbf{x}_j\|) \sim \varepsilon^{D_2}$$

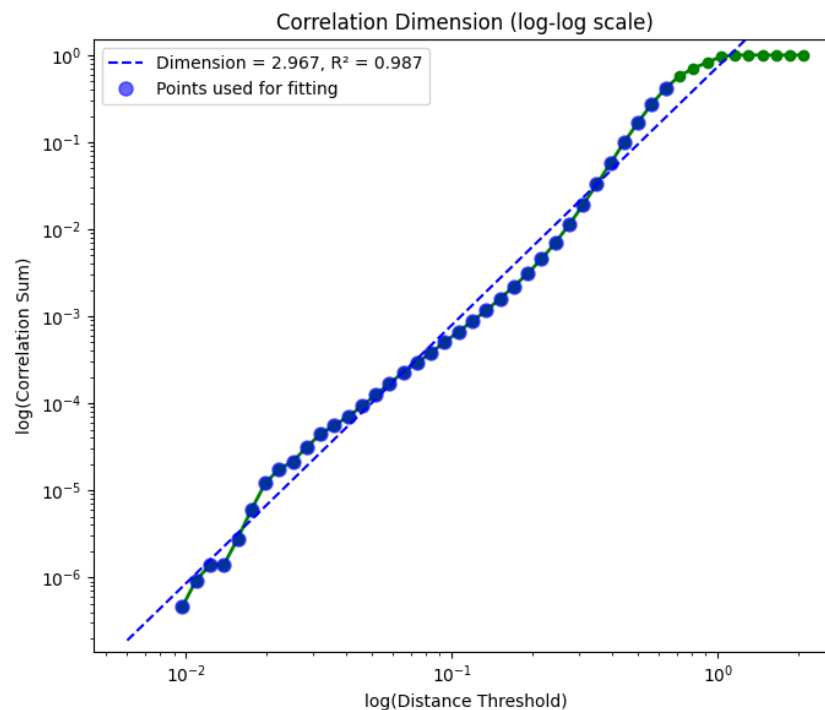


Typical chaotic system:
Power law relation



LLM

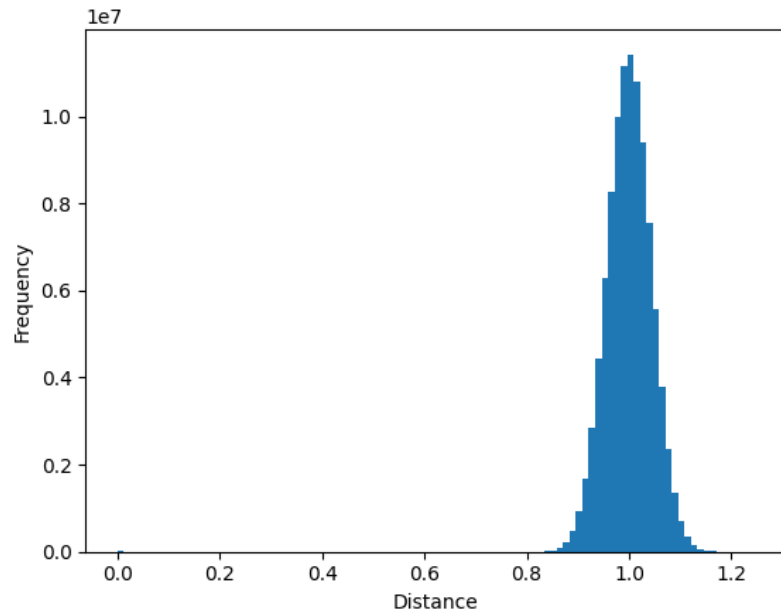
FIRST LAYER



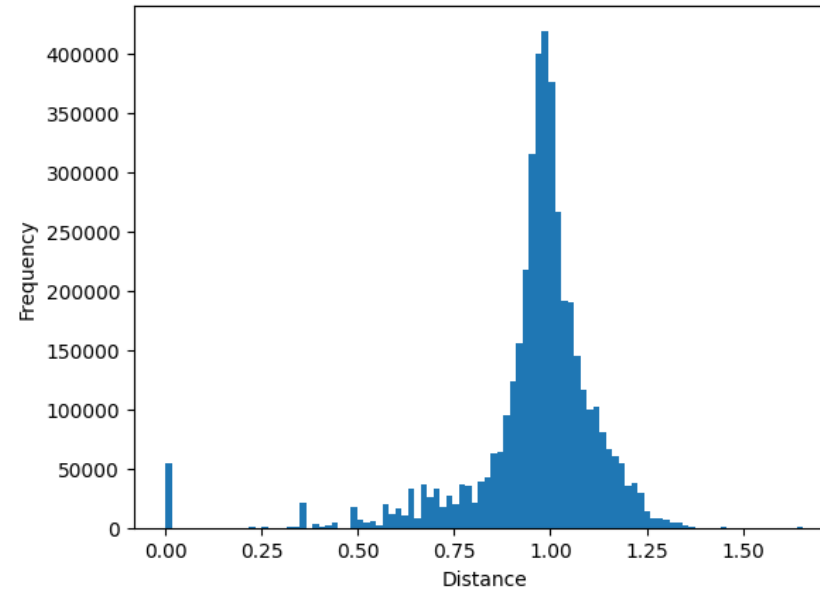
LAST LAYER

Distribution of Distance

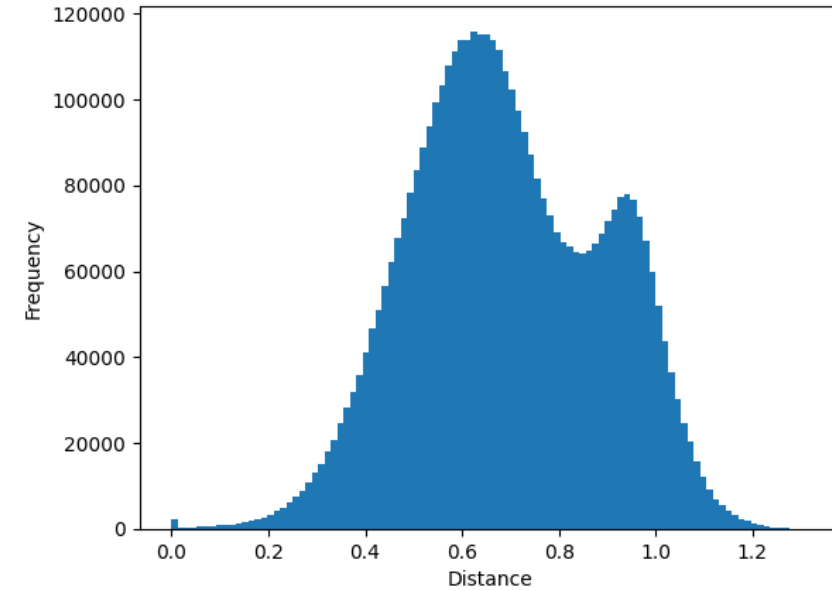
RANDOM



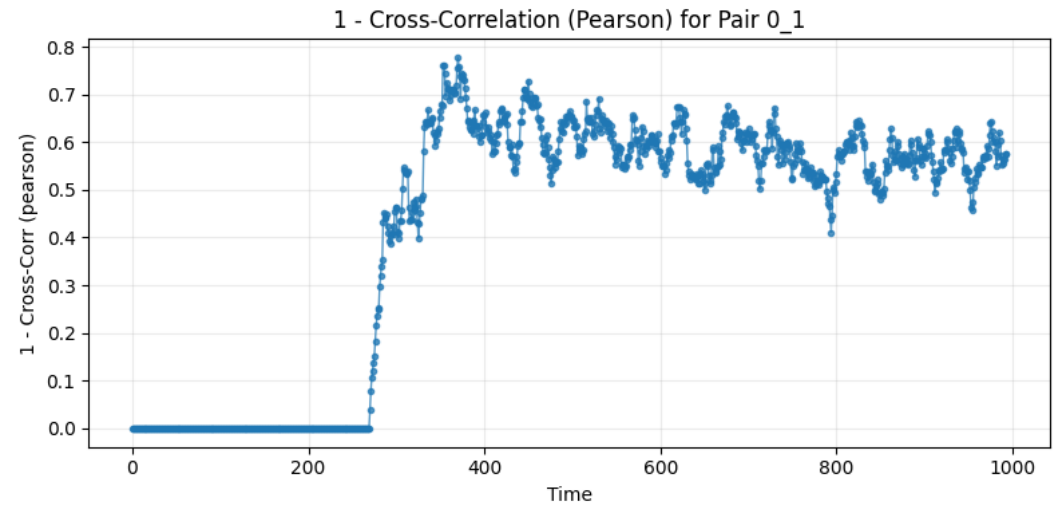
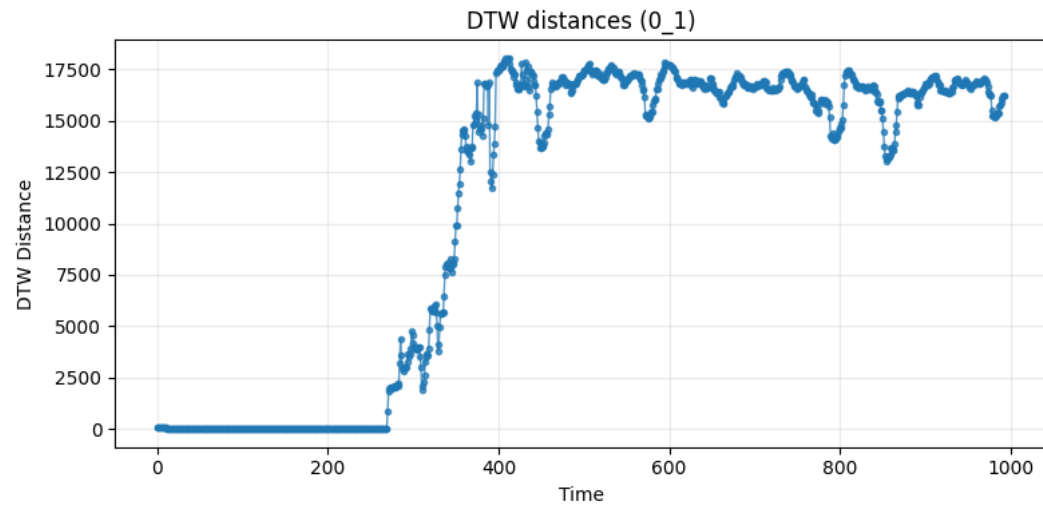
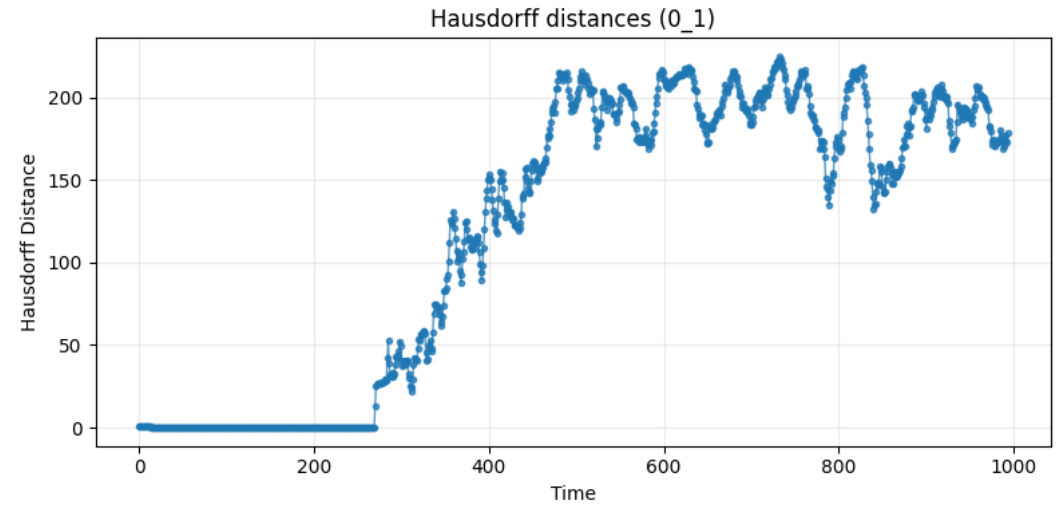
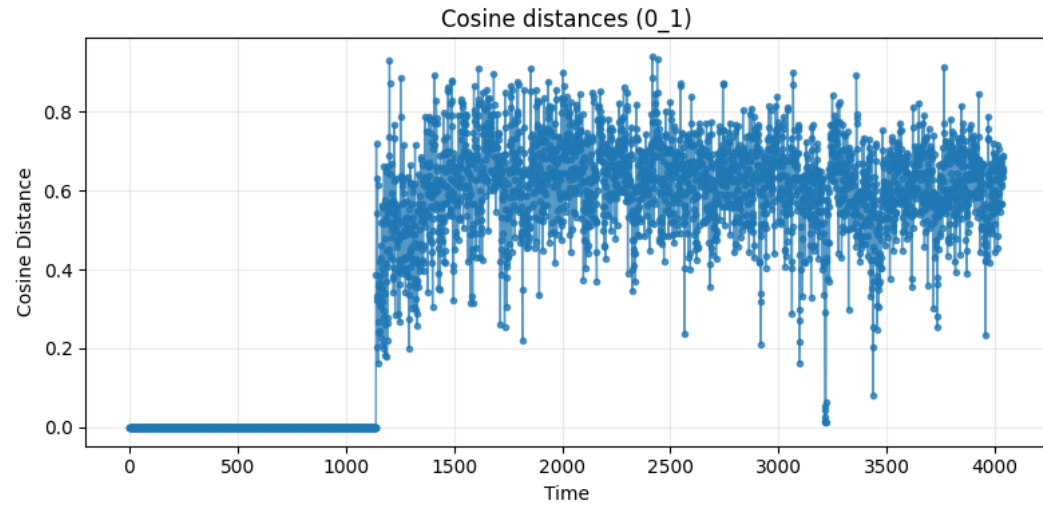
FIRST LAYER



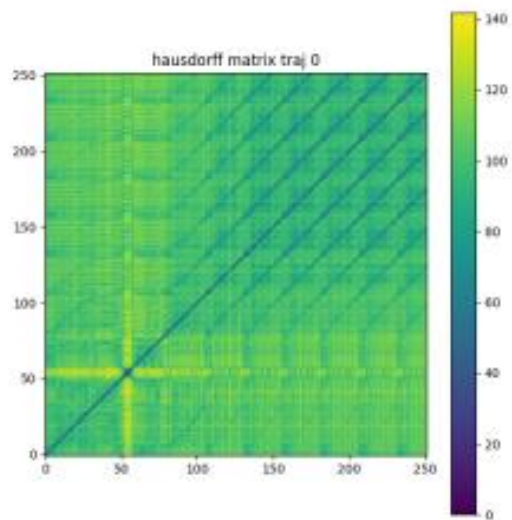
LAST LAYER



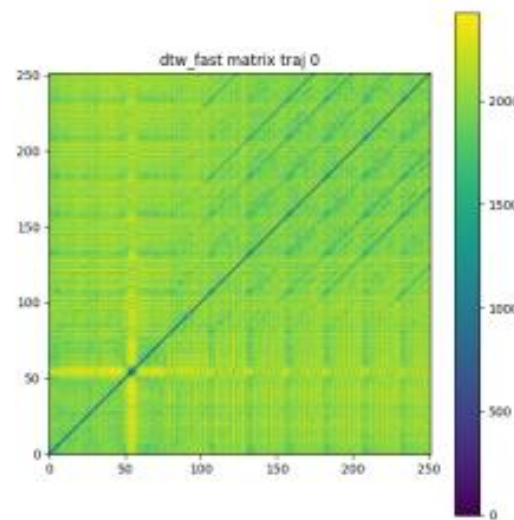
Alternative Distance Metrics:



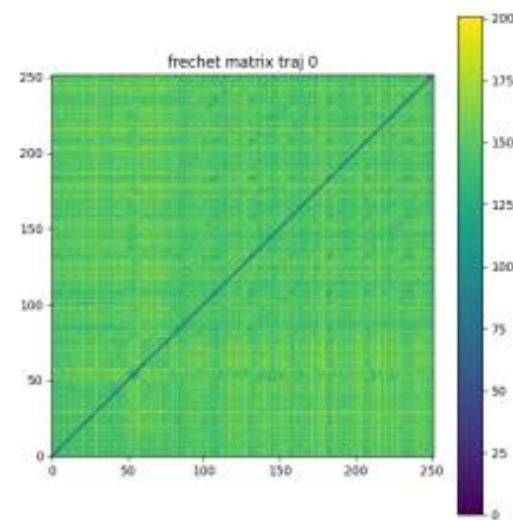
Distance Metrics: hidden states



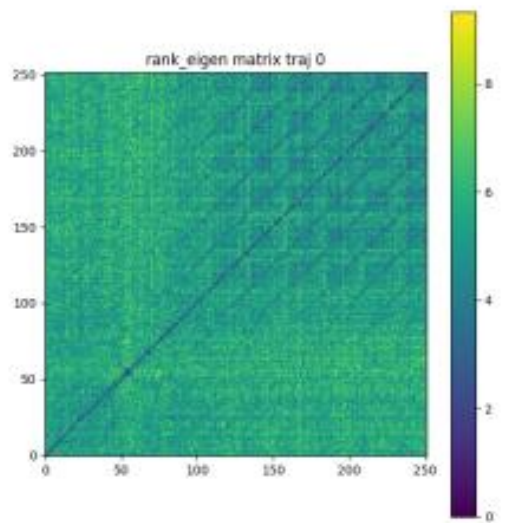
(a) Hausdorff



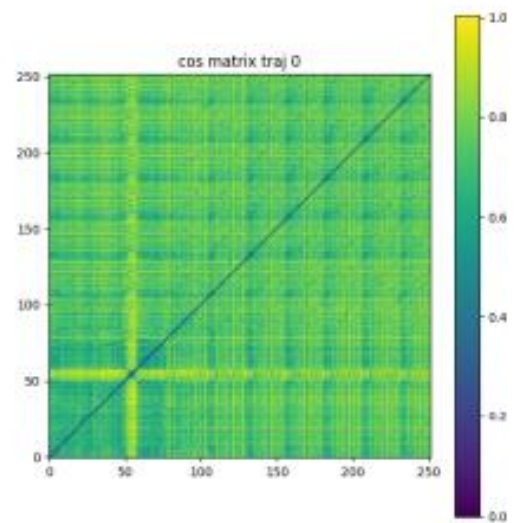
(b) DTW



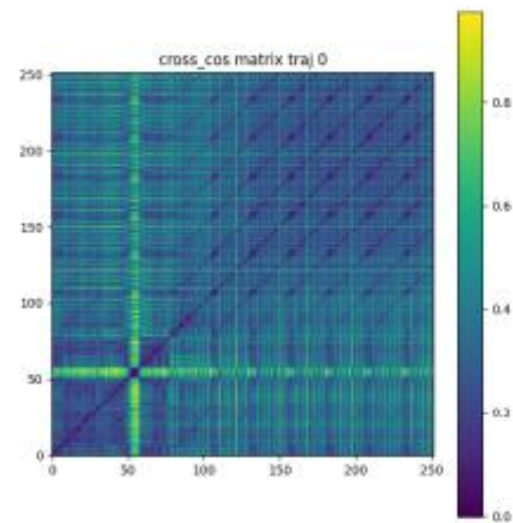
(c) Fréchet



(d) PCD

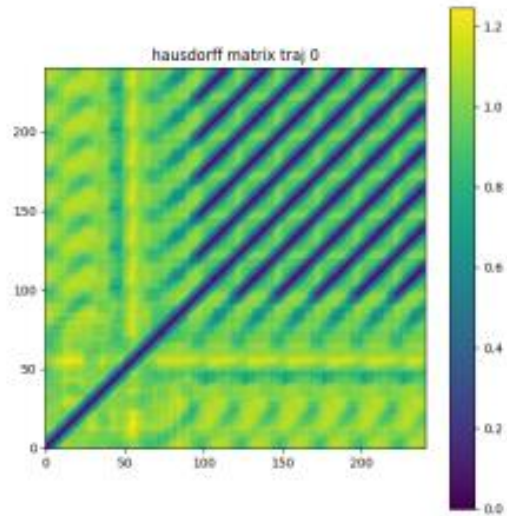


(e) Cosine Distance

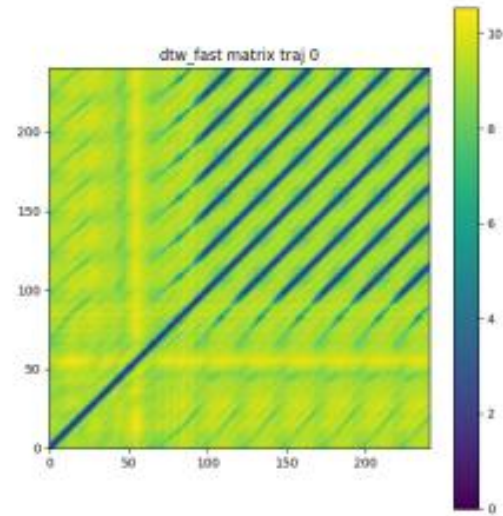


(f) Cross-Correlation Dissimilarity

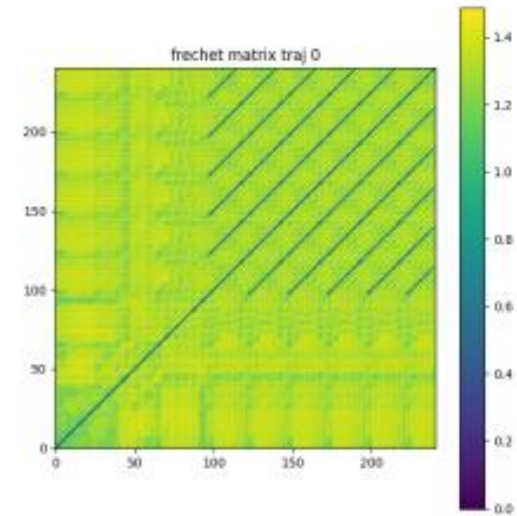
Distance Metrics: sentence embeddings



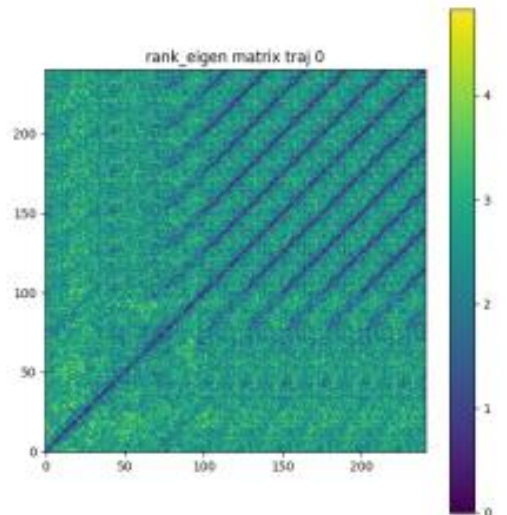
(a) Hausdorff



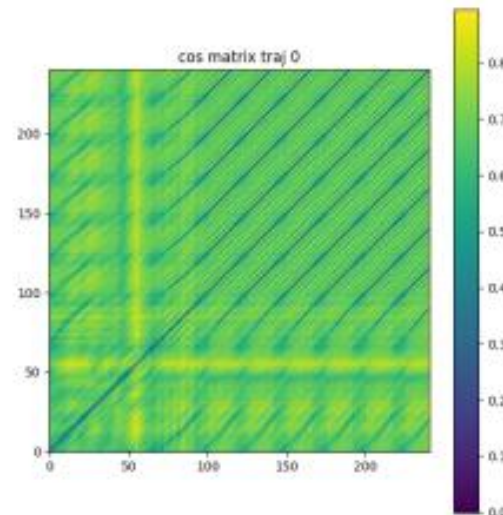
(b) DTW



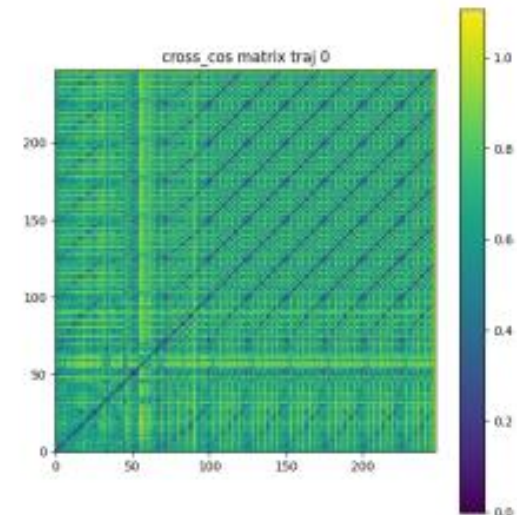
(c) Fréchet



(d) PCD



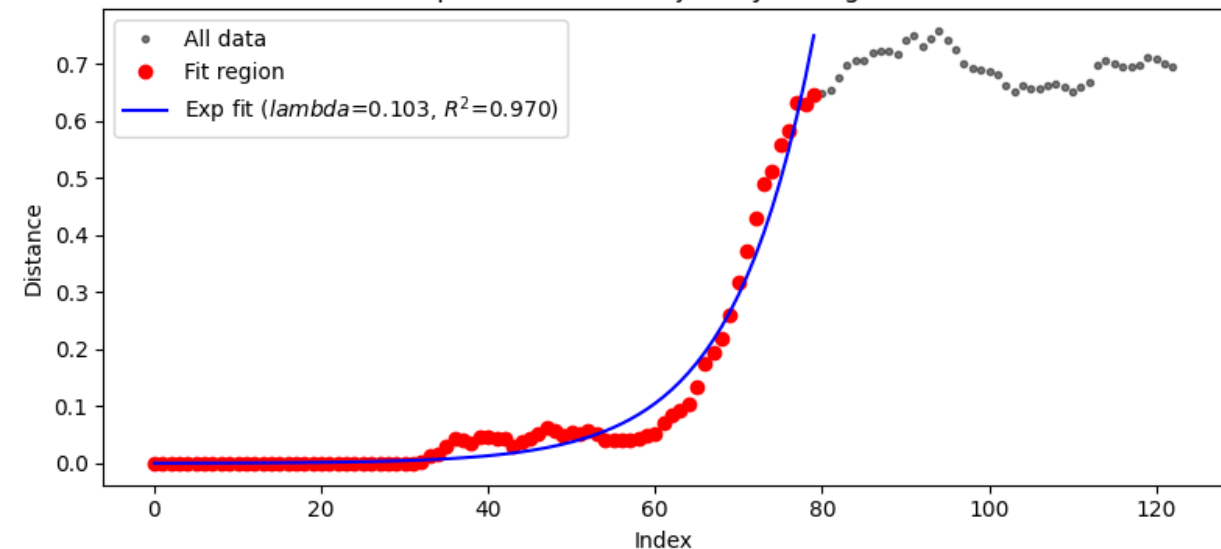
(e) Cosine Distance



(f) Cross-Correlation Dissimilarity

Sensitivity to initial conditions

Exponential Fit to Trajectory Divergence

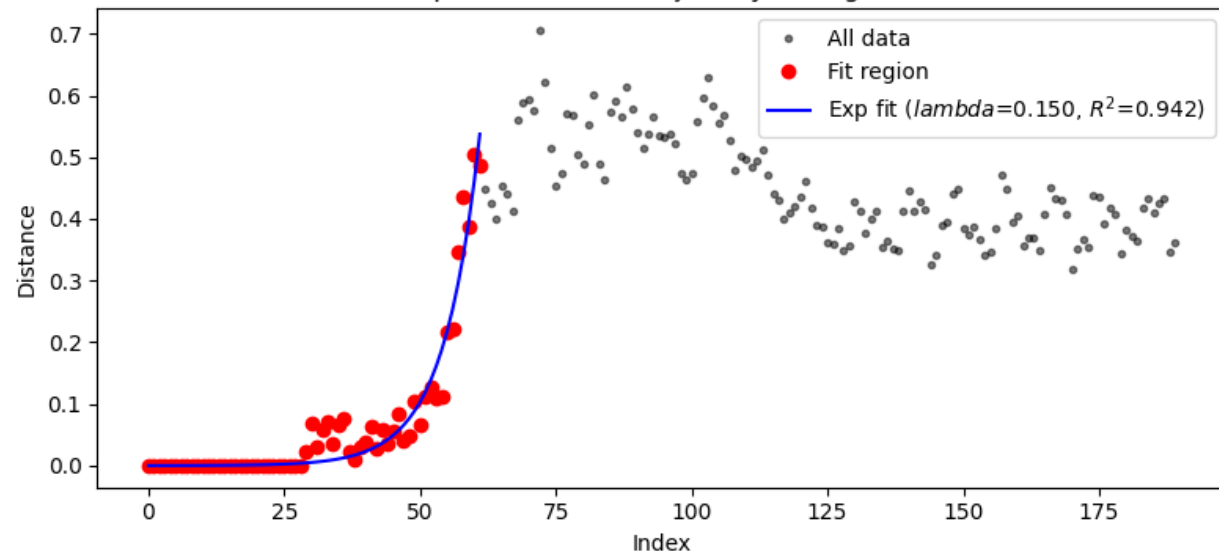


Sentence Embedding

Mean	0.16
Median	0.12
Std	0.079
Min	0.038
Max	0.29

Average r2 of the fits: 0.905

Exponential Fit to Trajectory Divergence



Hidden States

Mean	0.32
Median	0.20
Std	0.24
Min	0.06
Max	0.72

Average r2 of the fits: 0.97

Conclusions

LLMs exhibit chaotic behavior especially in the last layers.

Divergence and positive Lyapunov exponents indicate sensitivity to initial conditions.

Correlation dimension shows deterministic chaos.

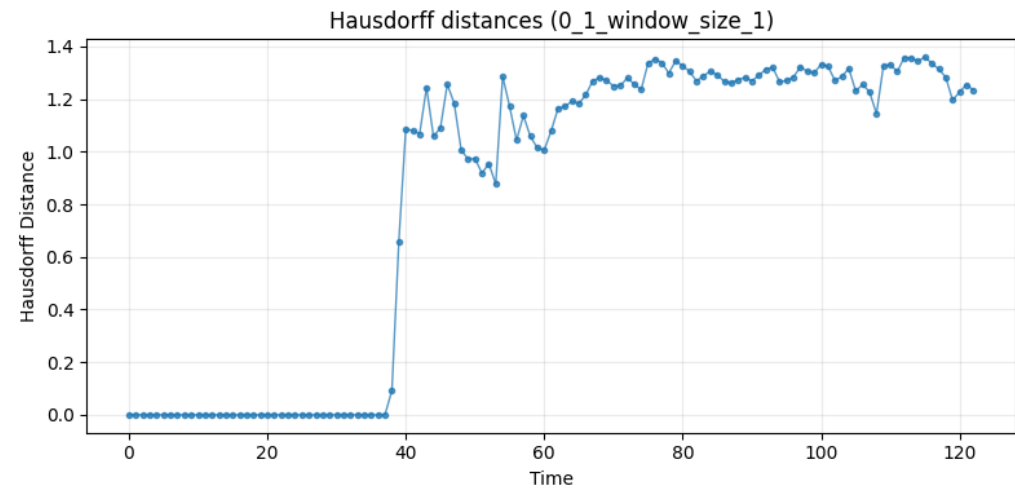
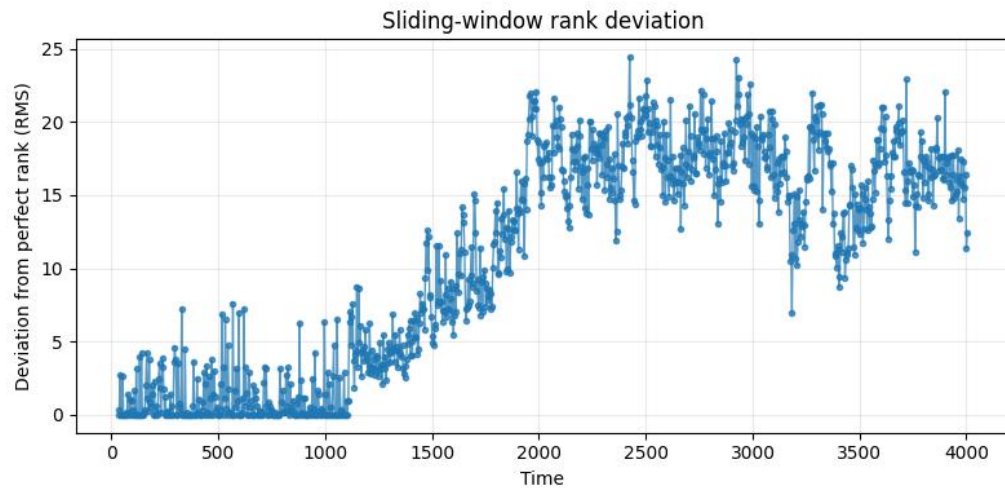
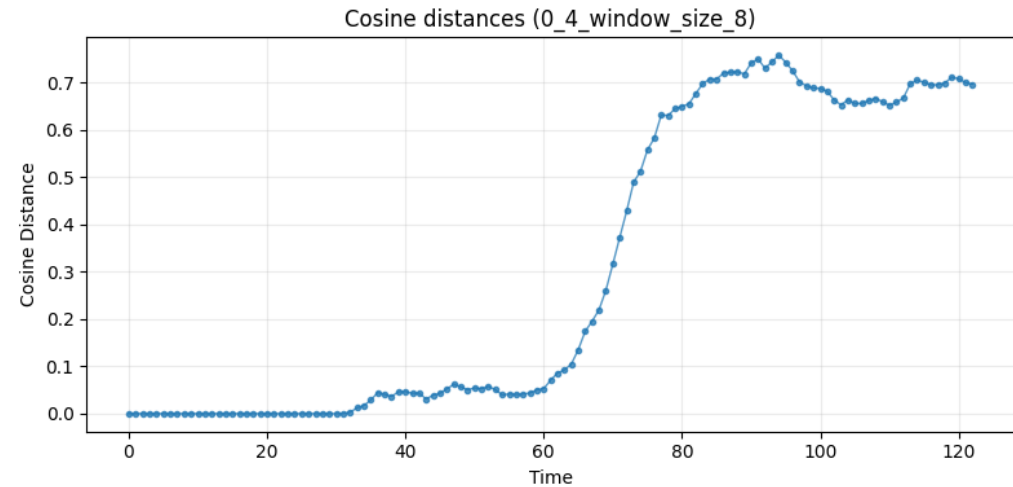
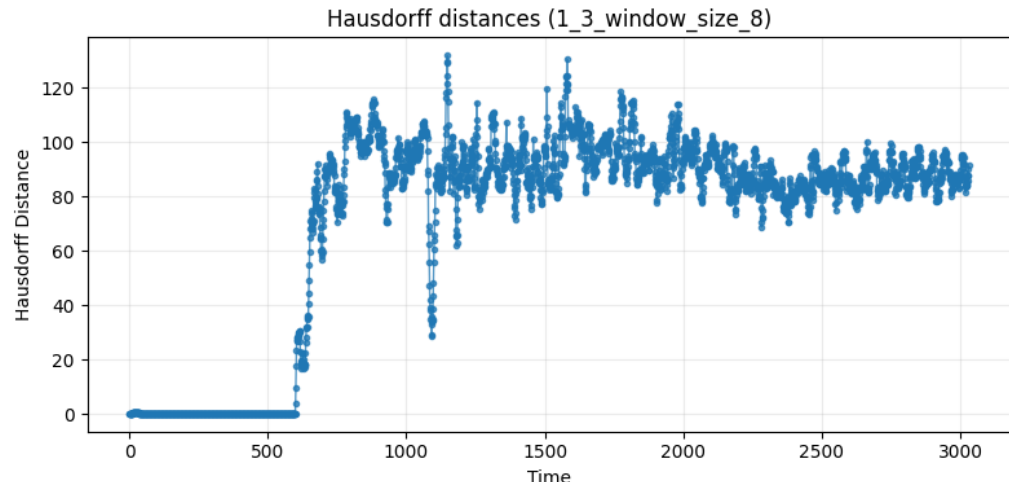
Recurrence Plots and RQA suggest similarity to other chaotic systems.

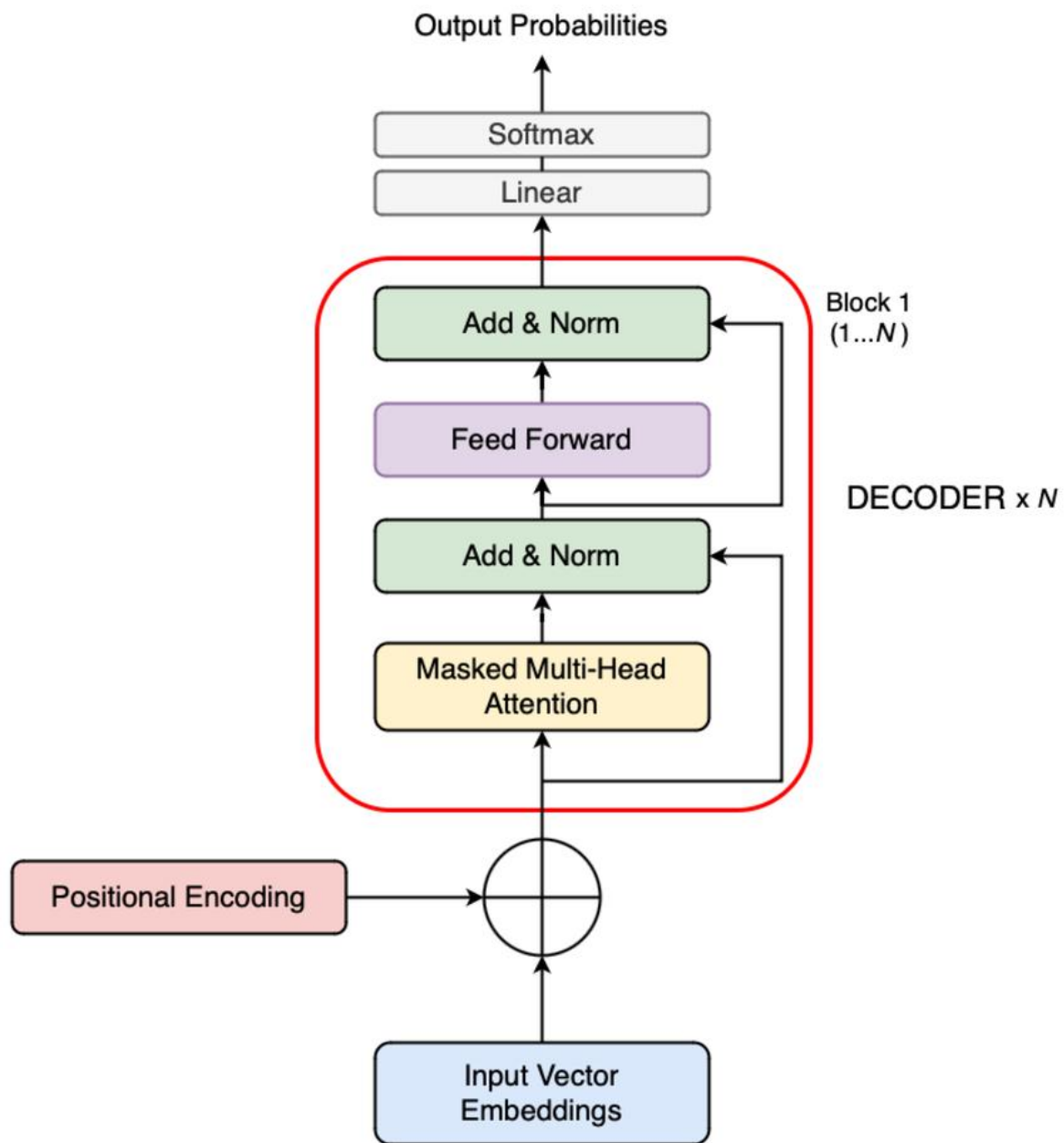
These results are consistent across distance metrics, and using both hidden states and sentence embeddings.

End.

I am extremely happy to answer questions!

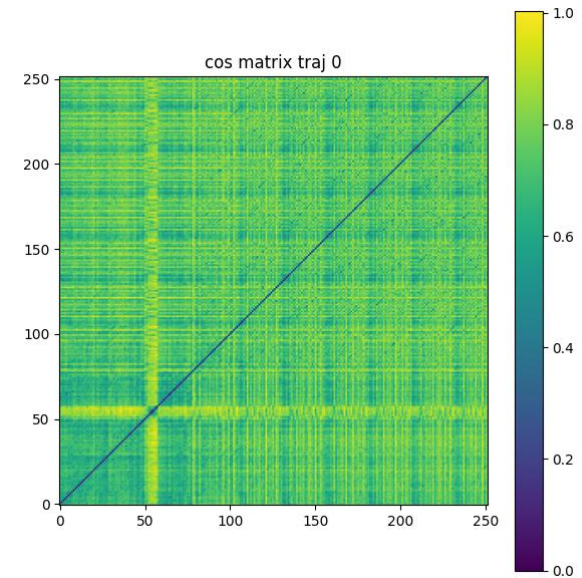
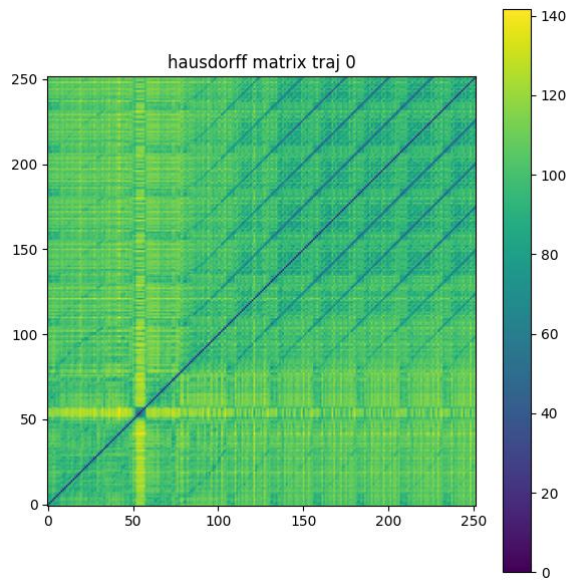
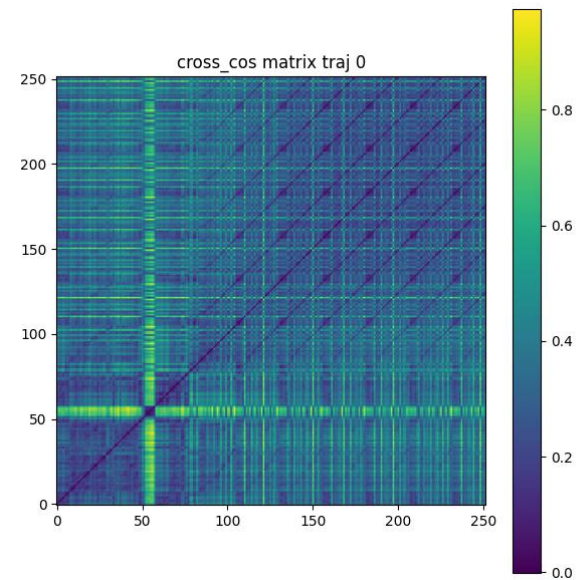
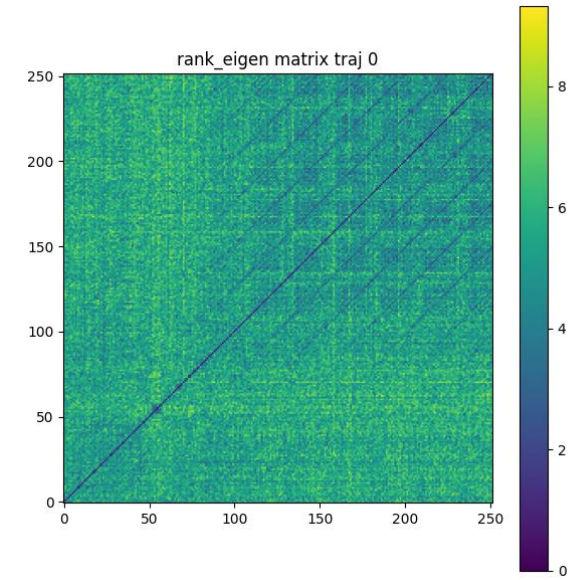
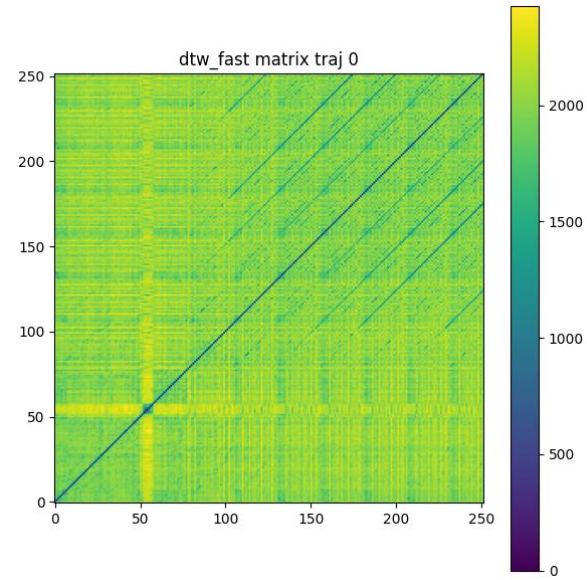
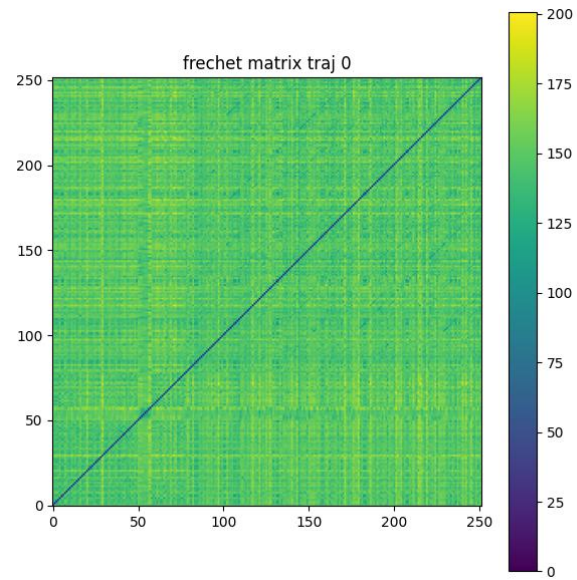
Divergence of trajectories



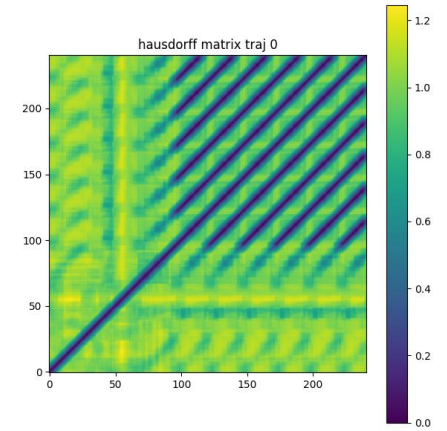
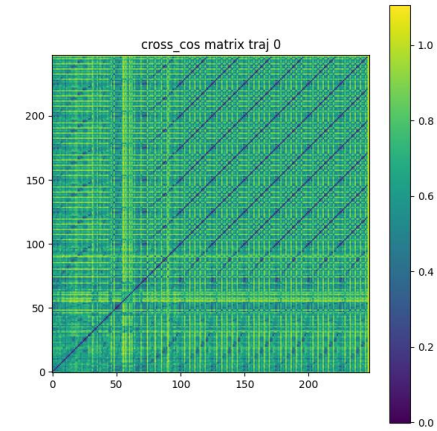
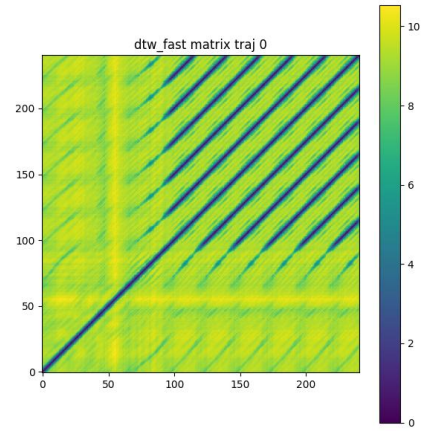
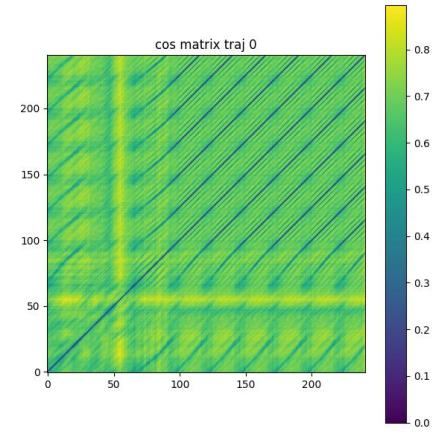
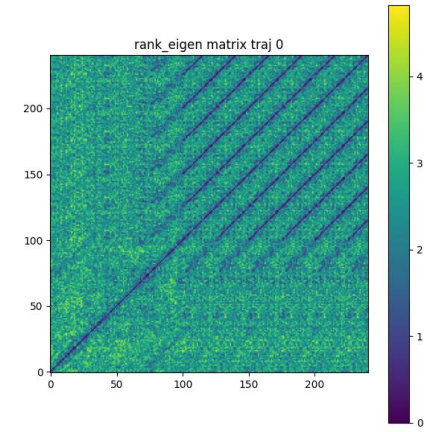
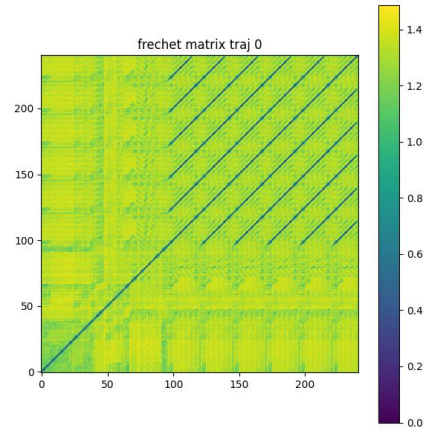


Chaotic system	LLM
Nonlinear coupling	attention
stretching	Attention + MLP
folding	Normalization layers + attention + add residual stream

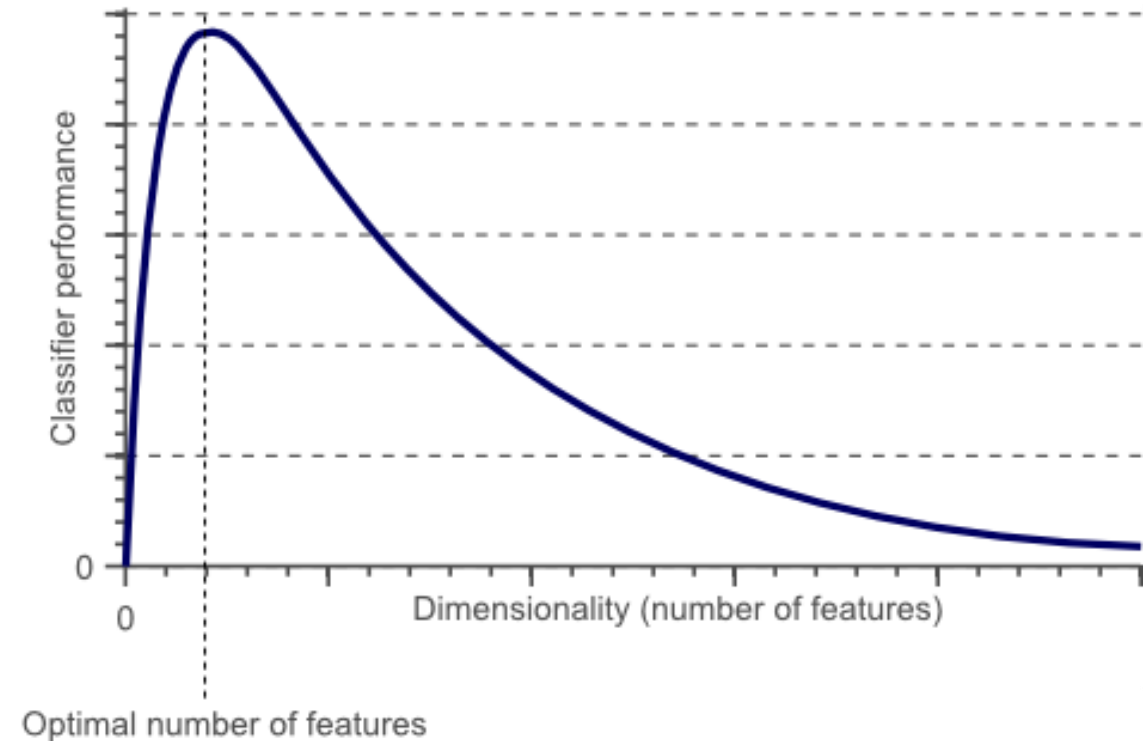
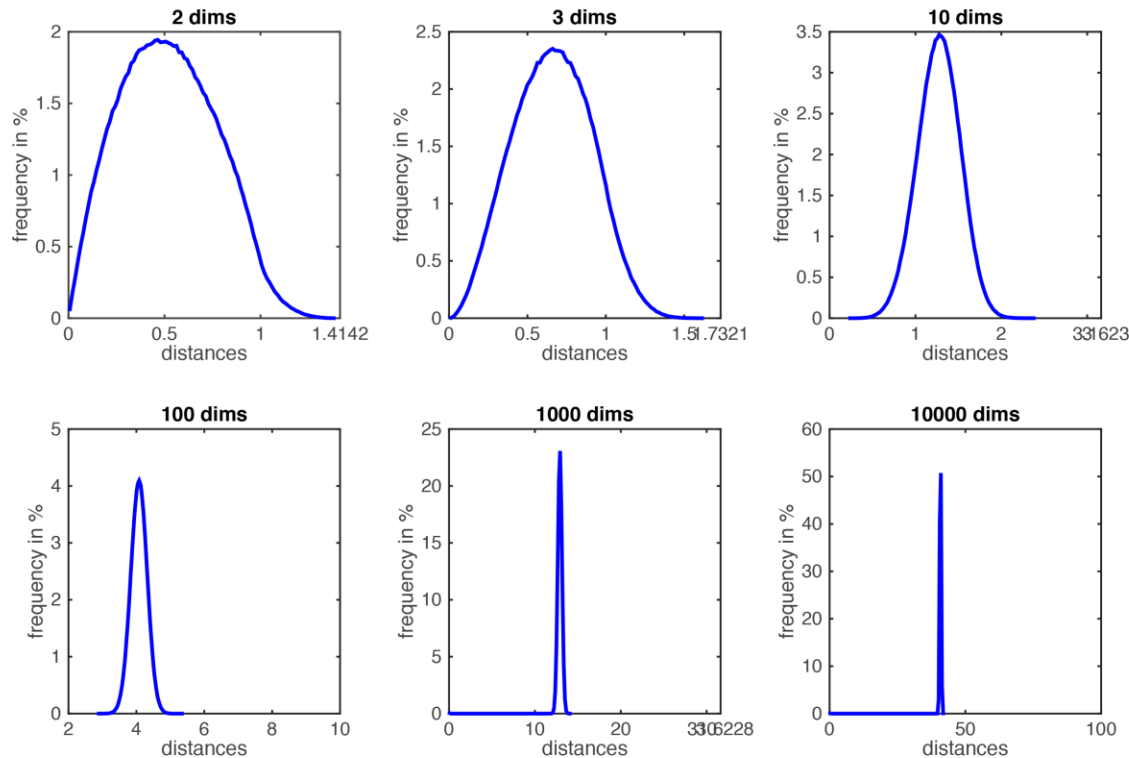
Distance Metrics



Distance Metrics

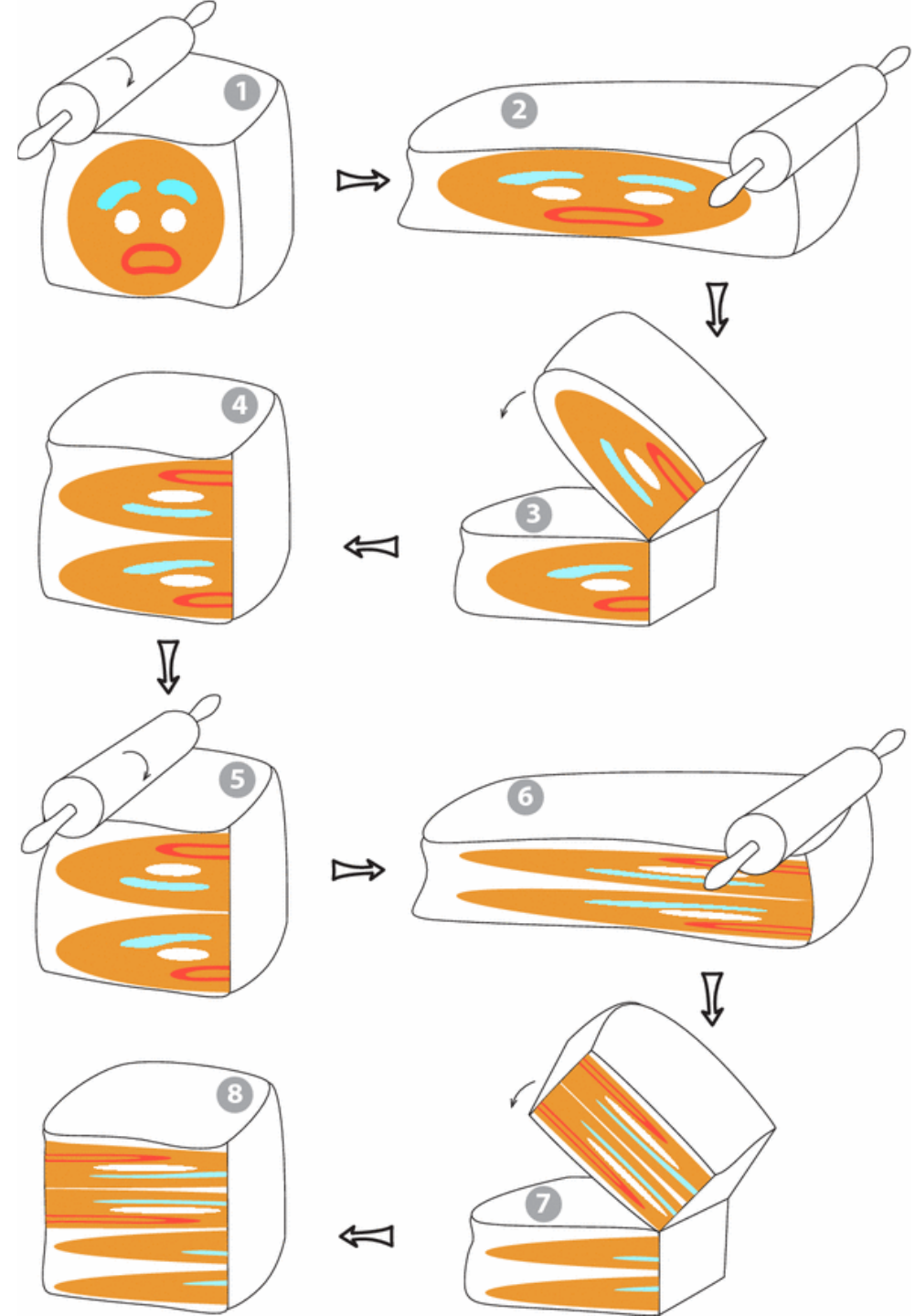


Curse of dimensionality: distance in high dimensions is complicated



Geometrical picture:

Stretching and Folding
drives Divergence and
Boundedness of chaotic
systems and strange
attractors



Chaos in LLMs

Gregorio Jaca 3rd year Physicist
engineering